Climate inequality

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Abstract

This paper explores the relationship between anthropogenic climate change and global inequality, a subject bridging two of the defining challenges of the 21st century but which remains virtually unstudied at subnational levels. To overcome a number of long-standing empirical limitations in the climate impacts literature, I apply recently developed methods for estimating dynamic responses to identified shocks new fine-scaled data on global distributions of income. First, I document new evidence that temperature shocks significantly and persistently impact distributions of income within countries, an effect driven by concentrations of harm onto the lowest income-earners in warm climates as well as a surprising vulnerability of the top 1% in these countries to environmental shocks. Next, I conduct a counterfactual analysis by integrating these inequality dynamics over observed distributions of income and the spatial incidence of global warming between 1980 and 2016. I show that climate change has regressively redistributed global income by depriving the world's poorest of economic opportunity that would otherwise have been available; for the poorest percentile of positive income-earners in our data, we find that income levels absent systematic warming since 1980 would be 29% $[18, 41]$ higher than realized. Further, I find that climate change over this period has reduced between-country inequality by 8.7% $[4.9, 13.3]$ and within-country inequality by 2.6% $[0.0, 5.6]$. Altogether, these results constitute the most comprehensive evidence yet of the regressive impact of climate change.

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1. Introduction

Mitigating anthropogenic climate change and narrowing economic inequalities are widely considered among the most pressing global challenges of the 21st century. For example, the Sustainable Development Goals adopted by all United Nations members explicitly resolve to "take urgent action to combat climate change and its impacts" and to "reduce inequality within and among countries" (United Nations, 2015). These global priorities are also understood to be intimately related, often framed as twinned crises (Pande, 2023): the Intergovernmental Panel on Climate Change identifies the "widening opportunities between and within countries" (IPCC, 2018) as first-order considerations in any sufficient policy response to climate change. Likewise, the World Inequality Report remarks that "climate mitigation is largely a distributional issue, not only between countries but also within them" (Chancel et al., 2022). When surveyed in 2021, an overwhelming majority of published environmental economists indicated a belief that climate change will "likely" or "very likely" exacerbate inequalities between (89%) and within (71%) countries (Howard & Sylvan, 2021).

These frames exemplify how the distributional implications of climate change are increasingly operationalized along its international and intranational dimensions. The conventional wisdom that the social consequences of climate change tend to concentrate in less developed countries with tropical geographies is by now supported by a robust empirical literature and reinforced by the visibility of unprecedented extreme weather events. While it is intuitive that an equivalent 'climate inequality' should also be reproduced *within* countries, such a distributional effect remains remarkably undercharacterized in the empirical literature even as intranational disparities comprise an increasingly dominant share of total global inequality[1] and public and institutional demand for these equity considerations has rapidly grown in recent years (Office of Management and Budget, 2023).

As a result, our understanding of how climate change affects the economic well-being of the Earth's eight billion human inhabitants is largely limited to aggregate comparisons across fewer than 200 polities and necessarily agnostic to urgent considerations such as social stratification and mass immiseration. Addressing this ignorance is necessary for adequately specifying the problem of climate change, let alone formulating an efficient and equitable response to it. This chapter aims to initiate and stimulate long overdue progress in this area by providing the most comprehensive characterization yet of the dynamic and distributional impact of climate change on economic inequality.

Credible progress in this area first requires overcoming long-standing challenges in the tradition of estimating the causal effect of weather variation on economic outcomes of interest (Pindyck, 2013). In Section 2, I briefly summarize specific technical issues in this niche literature to contextualize the methodological choices made in the present analysis. Perhaps the most pressing of these is that estimates of cumulative impacts implied by these "damage functions" are known to diverge by orders of magnitude depending on the modeler's assumptions about the persistence of these effects. While 76% of recently surveyed climate economists indicated a belief in highly persistent or permanent effects of weather on macroeconomic aggregates like GDP, statistical tests of their presence have been mostly inconclusive or heavily contested (Burke et al., 2015; Newell et al., 2021). Second, recent work has challenged the validity of raw weather variables conventionally used for as explanatory variables in these regressions on the grounds of identification issues. Finally, I discuss the use of these reduced-form damage functions to make inferences about systematic climate change.

My approach to addressing these concerns is described in Section 3 with more detailed exposition provided in Section A2. In short, I adapt recent innovations in empirical macroeconomics to construct more plausibly exogenous shocks and estimate dynamic impulse response functions by the method of local projections. To credibly measure global income distributions, my analysis uses new data from the most recent update to the World Inequality Database, which provides well-balanced percentile-year panels of pre-tax income for a set of countries now representing almost the entire global population. By applying a refined econometric apparatus to unprecedentedly rich distributional data, my empirical analysis accomplishes two objectives. First, it substantially improves the frontier of an analytic tradeoff between subnational disaggregation and global representativeness which has historically hindered the joint study of global inequality and climate change. Second, the dynamic estimates which result are parametrically 'bagge-free' in that they do not impose any restrictions on the persistence structure of these economic impacts, thus obviating the main source of extreme estimation instability observed in the climate impacts literature.

The results of this empirical analysis are documented in Section 3.2 where we present dynamic income distributional responses to identified temperature shocks. These novel representations depict how the expected social im-

[1]Within-country income inequality now accounts for almost 70% of total income inequality as measured by the entropy-based Theil measure of inequality, an increase from just over 40% in 1980. (Chancel et al., 2022)

pacts of temperature anomalies vary across income groups and climates and accumulate over time. We find evidence of substantial, significant, and persistent impacts on within-country distributions which are particularly pronounced in warmer countries. In these geographies, the distributional effect is driven by substantial weather-induced transfers of national income shares marked by severe reductions among the lowest income-earning decile and to a lesser extent, a perhaps surprising precipitous reduction in the share held by the highest-earning percentile. These losses are shown to compound for at least three years following a shock even when accounting for persistence in the shock itself. Because impacts to warmer countries are shown to be significantly negative across the entire income distribution, consistent with the prolific literature on country-level impacts, national income in sufficiently warm countries are expected to uniformly decrease in response to a transitory temperature shock. Thus, positive temperature shocks in warm countries are expected to regressively redistribute shares in a persistently diminished national income, acting as a drag on growth and widening existing inequalities.

In Section 4, we present a counterfactual analysis representing the most disaggregated evidence yet of the global economic impact of climate change. This entails integrating the distributional response surfaces estimated in the previous section over the observed histories of country-percentile-year income and the incidence of temperature shocks between 1980 and 2016. My results suggest that systematic changes to the distribution of temperature shocks over these 37 years has unambiguously worsened total global inequality, both between and within countries. This effect mainly stems from foregone income growth for the poorest 20% of global income-earners. For the lowest positive income-earning global percentile in our data, the analysis implies that incomes today would be 29% higher than observed had the climate remained stable at 1980 levels. The evidence is also consistent with the standard finding in the literature on aggregate impacts that climate change is extremely likely to have reduced total global income over this period though we cannot reject a null of no impact for segments of the global middle class, income groups largely comprised of individuals resident in cool climates where impacts have historically been relatively mild. Finally, I decompose these global distributional effects, finding that climate change over this period has increased between-country inequality by 8.7% [4.9, 13.3] and within-country inequality by 2.6% [0.0, 5.6] as measured by the mean logarithmic deviation index. To the best of my knowledge, the latter quantity has not been previously estimated.

Section 5 concludes by summarizing the contributions and limitations of the present analysis and suggesting promising extensions of this work. In par-

ticular, I briefly consider how the subject of climate-induced regressive redistribution of income poses urgent challenges the economic analysis of climate change and the design of climate policy.

1.1 Related literature

Spurred by methodological advances, accessibility of computing power, and political urgency, the past decade has seen a proliferation of economic research dedicated to estimating the social impacts of climate change. To contextualize the relative sparsity of within-country distributional impact studies in climate economics, it is instructive to familiarize the reader with the subfield's taxonomy of empirical designs into complementary "top-down" and "bottom-up" approaches each premised on an underlying utilitarian cost-benefit logic.

Top-down approaches restrict attention to a widely available social outcome of interest. This outcome is taken as a comprehensive enough proxy for social welfare that climatically attributable impacts to the stock of this outcome may represent the socially relevant climate "damage". The pioneering works in this category include Dell et al. (2012) and Burke et al. (2015), which relate annual observations of country-level temperature to GDP growth. Empirical designs in this category commonly take the form fixed effects regression models run on country-year panels. The advantage of the former approach is in its convenience and interpretability: the wide availability of cross-country economic aggregates such as those provided by national accounts substantially simplifies the task of calculating global climate impacts.

Alternatively, bottom-up approaches consider impacts to distinct "sectors" of an economy; mortality (T. Carleton et al., 2022), energy consumption (Rode et al., 2021), and labor (Graff Zivin & Neidell, 2014) are common areas of focus. Results using this approach can be presented as partial costs or combined with other sectoral damages in an integrated model which accounts for their interdependencies. The benefit of this approach is in its ability to account for disaggregated and non-market impacts to which single-dimensional aggregates are largely indifferent.

The simplicity of the top-down approach and comprehesiveness of the bottom-up approach pose a critical tradeoff between geographic disaggregation and global coverage. As perhaps the most prominent existing work directly concerned with characterizing global 'climate inequality', Diffenbaugh and Burke (2019) estimates that the ratio between the top and bottom deciles of the global income distribution was 25% larger in 2010 than it would have been had anthropogenic climate forcings been constrained to 1960 levels, all

else held equal. However, this quantity is calculated by assigning all individuals their country's mean GDP and crudely defining global deciles accordingly, an unavoidable restriction when limited to 160 or so distinct economic units. The authors remark that "documenting the impact of global warming on within-country inequality remains an important challenge."

S. Hsiang et al. (2017) is a particularly impressive example of a study which makes the much less common opposite compromise in this tradeoff. Integrating fine-scaled agricultural, crime, energy, mortality, and labor data, the authors determine that climate risks in the continental United States are disproportionately borne by counties in the US South and Midwest, regions generally already poorer than their coastal counterparts; climate change then is likely to imply worsening inequality within the United States. But to make more generalizable claims about the implications of global warming for global inequality, analyses must extend to settings which lack comparably high-quality and granular data. Elsewhere, Marx (2024) uses rich tax data from France to infer that years with marginally more days above 30°C are associated with wider income disparities between the richest and poorest cantons (administrative units described as roughly equivalent to 10 municipalities). Dasgupta et al. (2023) uses data from perhaps the most unequal country in the world to project a 3-6 Gini point increase in South Africa's income distribution by 2100 under a moderate warming scenario.

The distributional focus of this chapter complements a much more active area of research measuring unequal impacts across countries. Like Diffenbaugh and Burke (2019), a subset of these studies explore the implications of this type of inequality directly, for example by highlighting the inverse relationship between country-level contributions to global greenhouse gas emissions and exposure to their social consequences (Chancel & Piketty, 2015; Kotz et al., 2024; Ricke et al., 2018). There is by now general agreement that non-linearities in the response of aggregate variables to marginal warming concentrate damages and risk in relatively warm and poor countries which have contributed proportionally little to anthropogenic warming (Howard & Sylvan, 2021). However, estimated magnitudes of these impacts are notoriously divergent, largely because of unsettled disagreement over the persistence of these non-linear effects (Newell et al., 2021). We draw from recent work by Nath et al. (2024) and Bilal and Känzig (2024) which use less restrictive time series methods to circumvent these statistical bugbears.

Similar non-linearities emerge at the micro level across a range of outcomes as varied as labor supply (Graff Zivin & Neidell, 2014), agricultural yields (Schlenker & Roberts, 2009), and standardized test performance (Goodman et al., 2020). T. A. Carleton and Hsiang (2016) overviews the proliferation

of multidisciplinary studies in this category whose results describe patterns of setting-specific inequality suggestive of a broader micro-level environmental inequality similar to those observed in the aggregate. The present study bridges this micro evidence with the aforementioned macro evidence on inequalities between countries, estimating distributional responses consistent with the general features of both categories of impacts. We also document evidence of negative impacts to an elite minority, which I have not seen examined in existing work.

The literature on climate impacts on economic inequality *within* countries has remained comparatively sparse in large part due to an absence of adequate research infrastructure. Estimation of distributional equivalents of the aggregate damage functions standard in the literature requires data comparable in credibility and coverage to domestic production data whose construction across countries follows rigorous standards set forth by the international System of National Accounts (European Commission et al., 2008). Historically, distributional measurement of welfare has lacked and arguably been crowded out by the wider interest and investment in measuring aggregate production (Jorgenson, 2018). While secondary inequality data products compiling country-year inequality variable date at least as far back as Deininger and Squire (1996), the validity of these data products for cross-country causal analysis is limited by inconsistencies across primary sources in construction and availability (Atkinson & Brandolini, 2001). A recent empirical evaluation of the most widely used of these datasets concluded they are "not sufficiently credible" (Jenkins, 2015). In the absence of more credible global inequality measures, Cevik and Jalles (2023) and Gilli et al. (2024) estimate the model of Burke et al. (2015) replacing their GDP growth outcome variable with the inequality variables provided in these early-generation products.

The research programs of Thomas Piketty, Emmanuel Saez, and their coauthors since the turn of the century are widely credited with revitalizing public and academic interest in the measurement of within-country inequality. Beginning with work tracing the long-run evolution of income and wealth inequality in France (Piketty, 2003) and the United States (Piketty & Saez, 2003), their mobilization of extensive tax returns data has enabled credible measurement of top incomes over long time horizons, stimulating new programs of research and equipping the general public with new political language around which to mobilize (Jones, 2015). This paper takes advantage of the cross-country research apparatus that has rapidly developed and been made accessible in recent years with the intention of maturing the economic analysis of climate change in a similar manner. Palagi et al. (2022) is the only other climate-economic study I could locate which makes use of the same

"distributional national accounts" (World Inequality Lab, 2024) used in this study. The authors find that extreme precipitation negatively impacts low-income individuals much more severely in countries more dependent on agriculture.

Finally, I locate this paper in the context of a growing demand for distributional considerations in economic analyses of climate change and the design of climate policy. For example, Chancel et al. (2023) estimates that within-country inequality of carbon emissions have recently exceeded the equivalent "carbon inequality" between countries and advocates for targeted carbon taxation programs based on individual footprints to more efficiently internalize the climate change externality. The modern SCC framework accommodates equity considerations by concavifying the underlying utility function so that the welfare weights assigned to individuals are inversely related to their relative levels of consumption; a logarithmic specification is commonly chosen, for example. Variations of this more progressive utilitarianism have been institutionalized to some degree in a handful of countries including recently in the United States (Office of Management and Budget, 2023). Theoretically, the way equity weighting is implemented should result in substantially higher SCC estimates depending on the extent to which the distributive effect of climate change is modeled as regressive. This paper provides the first comprehensive estimate of the global regressivity of climate change at subnational resolution.

2. Data

2.1 Distributional income data

We make use of the recently published grouped income series from the World Inequality Database (World Inequality Lab, 2024). Its predecessor, the World Top Income Database launched in 2011, drew from comprehensive tax records from over 30 mostly Western countries to capture the distribution of income among their top income-earners with unprecedented precision. A critical innovation used to develop the new product is the systematic integration of this tax data with national accounts and household survey data in order to impute the full income distribution within those countries (Piketty et al., 2019) in a manner patterned after the System of National Accounts (European Commission et al., 2008). The newest generation of WID data extends coverage within those countries—now claiming to capture "100% of income" in the United States (Piketty et al., 2018)—and to countries with relatively limited available

data. By 2019, historical data for over 120 countries had been incorporated into the database. As of this writing, their equal-split adult pre-tax income series was available for 215 countries and territories collectively representing almost the entire global population with distributional data imputed every year since 1980. All income estimates are converted to 2023 US dollars using the provided PPP-adjusted exchange rates. Accompanying metadata scores the availability and reliability of each category of input source by country, unsurprisingly indicating that the input mix of these sources varies substantially by country with less developed countries being more reliant on household surveys and wealthier countries on tax records.

Notably for the percentile-level panels, the income data exhibits substantial bottom-coding with almost every country-year reporting exactly zero income for at least the bottom four percentiles of the pre-tax income distribution. Even if an accurate measure of adult unemployment, this evokes well-known deficiencies of income as an identifier and measure of poverty compared to alternatives like consumption expenditures (Meyer & Sullivan, 2003). For example, while the unemployed subpopulation of a given economy may well be composed of its most impoverished, it is likely to also include unemployed individuals of all levels of precarity and social status and does not account for security provided by communal and social safety nets. It also positively biases estimated changes in welfare among the poor since zero incomes mechanically cannot decrease.

Given these inherent disadvantages of income measures as a proxy for relevant welfare, we intentionally do not make precise claims about the poverty implications of our findings. Instead, we consider the incomes of the bottom 10 percentiles as a group in all analyses and consider estimated impacts for this quantile to be a conservative bound on true impacts on the very poorest because it implicitly assumes perfect equality among its constituent population (Cowell, 2011).

2.2 Historical weather data

We merge all income data with an annual country-level panel of population-weighted near-surface air temperature data derived from the Global Meteorological Forcing Dataset for Land Surface Modeling (GMFD), which corrects model-based biases in NCEP/NCAR reanalysis product using observational meteorological station data (Dept. of Civil and Environmental Engineering, Princeton University, 2006). The GMFD reports historical weather data for the

entire global surface at 0.25-degree resolution[2] for every three hours from the start of January 1, 1948 to the end of December 31, 2015. We use the product which reports daily averages of near-surface air temperature.

While other temperature series of similarly high resolution can provide a larger estimation sample by beginning before 1948 or updating more recently than 2015, results reported in the broad climate impacts literature have been generally consistent across data products. We favor the GMFD product as it has the unique feature of being the only reference product used to bias-correct and downscale the CMIP6 projection data series produced by the NASA Earth Exchange Global Daily Downscaled Projections; this allows the results of the present analysis to seamlessly integrate with future work exploring its implications under different future warming scenarios or attribution studies which attempt to decompose estimated climate change impacts into their anthropogenic and natural contributions, as introduced in Section A4.

2.3 Temperature as an explanatory variable

It has been customary in the climate impacts literature to use functions of observed weather variables such as temperature or precipitation as explanatory variables in causal analyses. Historically, their use in panel settings has been justified by an intuition that annual fluctuations in these geophysical conditions are plausibly exogenous with respect to most economic activity. This intuition has been increasingly challenged by works such as Nath et al. (2024), which argues that because levels of temperature do not satisfy conditions of an identified shock[3] and exhibit substantial autocorrelation, they are ill-suited for dynamic causal inference.

Addressing these concerns, I define temperature shocks $\widehat{\tau}_{it}$ as deviations in temperature levels relative to a climate 'state' variable $\overline{T}_{it}$. This state variable is in turn defined as the M-period moving average of local temperature $\overline{T}_{it}$:

[2] At this resolution, cells near the equator are approximately 27 km^2 (17 mi^2).

[3] As outlined in the handbook chapter of Ramey (2016), a valid shock in a time series setting must i) measure unanticipated movements in relevant exogenous variables, ii) be exogenous with respect to current and lagged endogenous variables, and iii) be uncorrelated with other exogenous shocks included in the model. Temperature specified in levels do not satisfy the first condition and the third condition is violated in commonly used non-linear specifications where temperature enters as a polynomial.

$$\overline{T}_{it} := \frac{1}{M}\sum_{m=1}^{M} T_{i,t-m} - 18 \tag{1a}$$

$$\widehat{\tau}_{it} := T_{it} - \overline{T}_{it} \tag{1b}$$

This definition of the state variable $\overline{T}_{it}$ can be contrasted against the time-invariant state variable used in Nath et al. (2024) where the climate state variable is constant for each country. I favor this moving-average construction because the median country in my sample observes, on average, a 0.26°C (IQR: 0.18-0.36) increase in this state variable every 15 years as a result of global warming trends. Additionally, this definition of the state variable has an appealing coherence with the convention in climatology of using long-run moving averages of weather realizations to define the slow-moving state of the climate at a fixed period. Most commonly, these "climate normals" are defined as 30-year averages to accommodate natural variation which would occur even in a stable climate[4] and so we set $M = 30$ in our main analysis; Kahn et al. (2021) is another economics study which uses the same construction.

We subtract 18 in (1a) so that the main effects we attribute to the temperature shock τ correspond to an intermediate representative climate of 18°C, roughly the 40th percentile of average temperatures in our country-year data.

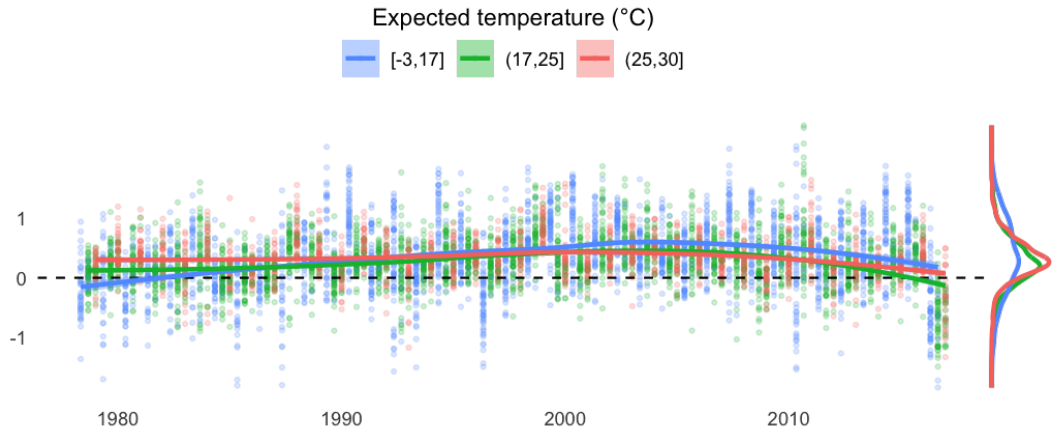
Figure 1 depicts the implied history of temperature shocks $\widehat{\tau}_{it}$ constructed in this manner with individual points corresponding to distinct country-year observations. These observations are divided into three equally sized groups according to their corresponding climate states $\overline{T}_{it}$; points and local regression curves are colored accordingly. The marginal distribution plots on the right demonstrate that colder seasonal countries experience more positive and more volatile temperature shocks than their warmer counterparts, a reflection of both greater natural weather variability in these settings and the fact that absolute anthropogenic warming is greater in regions closer to the Earth's poles[5].

The positive values of the curves in Figure 1 over almost the entire observation period captures the idea that systematic warming is characterized by

[4]The most economically relevant example of this natural variability is the El Niño Southern Oscillation, an ocean-warming phenomenon which can vary average global surface temperatures by as much as 0.4°C across its three phase cycles which may span 2-7 years. Other minor oscillations can span multiple decades.

[5]Figure A5 provides a more direct illustration of this phenomenon for different global climate models

Figure 1: Country-level time series of identified temperature shocks



Points correspond to shocks observed in individual country-years. The three colors correspond to an equal-sized binning of country-years according to their moving-average temperature. Flexible local regression curves describe time trends for each bin; segments with more positive values correspond to periods of greater systematic warming. Marginal distribution plots on the right show that cooler country-years observe greater variance in shocks and more positive values.

the increased frequency and greater magnitude of positive (hot) shocks relative to negative (cold) shocks. Intervals where the curves are closer to 0 thus reflect relative balance in the incidence of hot and cold shocks, corresponding to periods of relative climate stability and slower warming.

2.4 Modeling the persistence of social impacts: levels vs. growth effects

Comparisons across columns in Figure 2 illustrate the conceptual distinction between a shock which imposes transitory level effects (first column) and permanent growth effects (third column); the middle column describes a middle ground where an effect is persistent but eventually vanishing. Effects are measured in logged levels (top panels) and first logged differences (bottom), a convenient approximation of a growth rate. Filled lines depict the evolution of the outcome variable in the absence (black) and presence (red) of a unit shock which occurs in period 0. Filled areas in orange measure the cumulative loss (or gain) in the outcome which may be attributed to the shock.

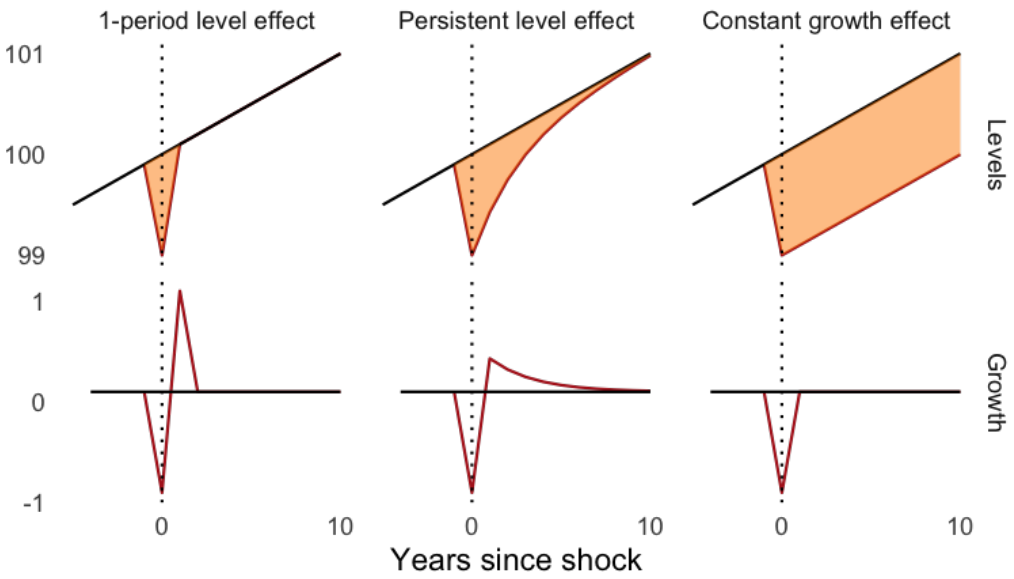
Under a one-period level effect, the level of, say, GDP falls by an estimated level $\widehat{\beta}_0$ in the year of the shock but fully recovers to its pre-shock trend (black) in the next period. In this scenario, the production lost in the year of the shock is exactly offset by an above-trend rate of growth which exactly offsets the negative growth effect in the year of the shock. The shock has no additional effect on future production and the area in orange represents the one-off impact from the shock.

Under the constant growth effect depicted in the rightmost panels, the same shock incurs the same effect on the levels but this effect is permanent so that GDP in all future periods is reduced by the same amount relative to the no-shock counterfactual. The difference between the two trajectories thus compound over time as observed by the orange area which grows linearly with the projection horizon. The corresponding growth effect depicted in the bottom panel is entirely transitory with no compensatory positive growth relative to trend.

The middle column describes an intermediate case where the economy requires several periods to return to its pre-shock trend. Cumulative differences attributable to the one-off shock continue to rise over time but to a diminishing degree until eventually compensatory positive growth exactly offsets the initial negative growth, i.e., the integral of the growth curve equals zero in the long run.

Now consider the following two models, which we may refer to as the levels and growth models respectively.

Figure 2: Levels and growth under different persistence structures



Illustrative diagrams depicting annual levels (top row) and growth (bottom row) of a social variable of interest. Dotted vertical lines mark the timing of a transitory temperature shock. Red lines represent trajectories resulting from a transitory temperature shock while black filled lines represent their counterfactual trajectory had they continued their pre-shock linear trend. Filled areas in orange correspond to the implied cumulative loss in the outcome variable ("damages") attributable to the shock. Columns correspond to different assumptions about how persistent the impact of a temperature shock is on the social outcome.

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$$\log Y_{it} = \beta^L X_{it} + v_{it} \tag{2a}$$
$$\Delta \log Y_{it} = \beta^G X_{it} + u_{it} \tag{2b}$$

For simplicity, imagine the outcome variable Y_{it} represents a conventional economic aggregate like GDP and X_{it} a perfectly identified temperature shock. In the levels model, β^L represents an estimate of the change in log GDP attributable to a unit shock. Graphically, this impact is represented by the vertical distance between the black and red lines in the top-left plot. Since this is an entirely static model, there is no mechansim for X_{it} to affect future values of Y_{it} and so the model assumes a perfect return to trend in all ensuing periods. In the growth plot directly below, this is represented by a $\beta^L - 0 = \beta^L$ effect in period 0, an exactly offsetting $0 - \beta^L = -\beta^L$ effect on growth in period 1, and no effect on growth in all ensuing periods. In models specified in levels, level effects which persist for p periods can be accommodated by including p additional lags of X in the regression. A permanent growth effect can only be accommodated with the inclusion of infinite lags.

In the growth model, $\beta^G \neq 0$ represents a permanent change in the difference between consecutive values of Y_{it}. This is represented by the gap between the black pre-shock trend line and the observed GDP series in the top-right plot staying constant at β^G for all periods following the shock and is never offset. Level effects which persist for p periods can be accommodated by including p lags of X and finding that the sum of the $p + 1$ coefficients is exactly 0 so that the initial negative growth effect is eventually offset by sufficient compensatory growth.

The differing restrictions imposed by these stylized regression models is a simplification of perhaps the most influential unsettled question in climate economics: the persistence of climate impacts. For example, two studies projecting the proportional loss in cumulative global production by 2100 arrive at point estimates of -20% (Burke et al., 2015) and -3.4% (Casey et al., 2023) under the same worst-case warming scenario. Methodologically, the two studies use similar methods and data except for their choice of outcome variable.

Intermediate persistence structures where level effects are diminishingly persistent can be accommodated by either model by including sufficient lags of the explanatory variables. Of course, identifying variation is reduced with the addition of additional explanatory variables, a problem exacerbated in specifications where shocks enter non-linearly and settings where shocks exhibit significant autocorrelation, both general features of modern climate models using absolute temperature as their explanatory variables. In addition,

models specified in levels are notoriously vulnerable to unit root concerns, further undermining credible identificaiton.

Indeed, the appendix to Burke et al. (2015) concludes that the identifying variation which remains after lag augmentation to test for persistence "cannot reject the hypothesis that this effect is a true growth effect nor the hypothesis that it is a temporary level effect", rendering these extremely influential assumptions effectively unfalsifiable. Absent substantially improved data, overcoming this issue requires richer data than provided by national accounts or an alternative methodological approach relatively robust to misspecification.

3. Empirical analysis

3.1 Methods: cumulative temperature multipliers

To characterize the dynamic effects of an identified temperature shock on national income distributions, we adapt the method of local projections first introduced in Jordà (2005). As Jordà (2023) notes, this semi-parametric method of estimating impulse response functions has found growing appeal outside of applied macroeconomics in recent years as recent econometric literature has demonstrated its flexibility in incorporating non-linearities (Cloyne et al., 2023), its coherence with the potential outcomes framework (Dube et al., 2024), and its robustness to misspecification and non-stationarity (Montiel Olea & Plagborg-Møller, 2021; Montiel Olea et al., 2024; Piger & Stockwell, 2023). Berg et al. (2024), Bilal and Känzig (2024), and Nath et al. (2024) are the first works I am aware of to apply this method to estimate dynamic economic responses to climate shocks, demonstrating how these features can help address the identification issues described in Section 2.

Our application of this method to our inequality setting entails the following steps:

1. Estimating the state-dependent response of income to a unit temperature shock
2. Estimating the state-dependent response of temperature shocks to a unit temperature shock
3. Using both to estimate a state-dependent multiplier to isolate the dynamic income response net of the shock's effect on future shocks
4. Statistically evaluating the significance and persistence of this multiplier

5. Repeating the above steps for each income quantile of interest

We briefly describe this process below and provide additional exposition in Section A2 using an application to country-level GDP data.

3.1.1 Estimating state-dependent impulse responses

We use country-year panel datasets to estimate dynamic responses to temperature for an outcome of interest represented by the outcome variable y_{it}. For example, y may represent the average logarithm of income of the bottom 10% of individuals in country i in year t and we may repeat the estimation process for each income decile and use the results to infer a total effect on national income distributions.

The method of local projections entails estimating $H+1$ iterations of the following single-equation model each corresponding to different horizons $h \in \{0, 1, ..., H\}$ periods after an identified temperature shock:

$$\begin{aligned} \Delta_h y_{i,t+h} &:= y_{i,t+h} - y_{i,t-1} \\ &= \beta_{1h}\widehat{\tau}_{it} + \beta_{2h}\widehat{\tau}_{it} \cdot \overline{T}_{it} + \lambda_h \overline{T}_{it} + \boldsymbol{\gamma}_h^\top \mathbf{Z}_{it} + \mu_i + \eta_t + u_{i,t+h} \end{aligned} \tag{3}$$

In our main specification, the matrix $\mathbf{Z}_{it}$ includes lagged controls $\{\widehat{\tau}_{i,t-j}, \widehat{\tau}_{i,t-j} \cdot \overline{T}_{it}, \Delta y_{i,t-j}\}_{j=1}^{p}$. The inclusion of p lags of the temperature shock $\widehat{\tau}$ is included to address potential autocorrelation in the explanatory variable. Interactions of these lagged shocks with the state variable capture potential state-dependencies in these autocovariance structures motivated partially by the finding in Figure 1 that shocks in cooler climates tend to be more positive on average and feature greater year-to-year variability. Lags of the outcome are included as a guard against trend non-stationarity. Finally, the sets μ_i and η_t of fixed effects are included to absorb time-invariant unit-specific effects and common year-specific effects. In all, this general model differs from the state-dependent model of Nath et al. (2024) only in how temperature shocks are constructed and in permitting the state variable to vary over time as described in Section 2.3.

For each projection horizon h, the main regression coefficients of interest attribute to a unit temperature shock a main effect β_{1h} representing a change in the "long difference" between y observed h periods after the shock and y observed the period before the shock. Equivalently, this can be interpreted as the change in y h periods after the shock relative to a counterfactual absent the shock. In most of our results, y represents the logarithm of income for

some quantile of the national income distribution; in these cases, coefficients are interpreted as percentage-point impacts on income.

An interaction term β_{2h} allows the effect at each period to scale with an observation's time-varying expected temperature $\overline{T}_{it}$, defined in Section 2.3. This 'state dependence' is motivated by the standard result in the climate impacts literature that the marginal effect of a given temperature shock on aggregate economic outcomes is increasingly negative for warmer climates.

The inclusion of a main effect λ_h associated with the time-varying state variable allows for the possibility that expected impacts may depend on changes in expected temperatures, net of the impact of climate change on the distribution of temperature shocks. For example, if a hot country stabilizes at a climate that is 1°C warmer than some reference period, the productive potential of the country is altered even in the absence of temperature shocks because the new climate will be more or less conducive to different economic activities. Neglecting this likelihood amounts to assuming perfect climate adaptation in the long run and that climate change is only socially impactful through its positively biasing the distribution of annual temperature shocks.

It is worthwhile to take stock of the assumptions underlying the sequence of regressions specified in (3). First, because coefficients for each horizon are estimated through distinct regressions using differently constructed variables from different subsamples of the data, there are in principle no restrictions imposed by the model on the shape of the resulting impulse response function for a given value of the state variable; it is in this sense that the method is partially non-parametric.

This flexibility is particularly appealing in the context of estimating responses to weather shocks. As summarized in Section 2.4, fully parametric methods require the modeler to assume either of the extreme persistence patterns distinguished in Figure 2 with limited ability to accommodate intermediate persistence dynamics. By instead allowing the modeler to remain mostly agnostic to the data generating process, this method circumvents a notorious source of extreme estimation instability and can accommodate any persistence structure in principle without restriction[6]. Graphically, the sequence of coefficient estimates $\{\beta_{1h}\}_{h=0}^{H}$ in (3) maps directly to the differences between the black and red time series in the first "levels" row; equivalently the sequence defined as the first difference in this sequence of coefficients correspond to the difference between the red and black time series depicted in the second "growth" row. Thus, a one-period level effect would be represented by a sequence where $\beta_{1h} = 0$ for all horizons $h > 0$ while a constant growth effect

would imply a sequence converging to a non-zero constant[7].

The single-equation local projections model is asymptotically equivalent regardless of whether the outcome is specified as h-period long differences as in (3) or as h-period projections in levels. However, recent work has shown that the former has particularly appealing finite-sample properties which make it especially robust to misspecification, largely eliminating well-known biases associated with highly persistent and even non-stationary processes (Jordà, 2024; Piger & Stockwell, 2023). This is particularly valuable in a climate impacts literature where GDP regressions are commonplace despite their vulnerability to unit root concerns (Campbell & Mankiw, 1987; Nelson & Plosser, 1982).

Still, the model is not entirely free of restrictions. Most critically, the model specified in (3) imposes within-horizon linearity assumptions on the coefficients of interest. Since the shock enters additively into (3), responses to the shock are assumed to scale proportionally so that a 1°C shock has exactly twice the impact of a 0.5° shock and the exact negative of the effect of a -1°C shock. By the same principle, the model also imposes linearity on the interaction effect, imposing symmetry in the state dependence such that holding all other factors fixed, the expected impact of a shock is assumed to be $k\beta_{2h}$ percentage points more positive in a climate some quantity k degrees warmer than baseline and $-k\beta_{2h}$ more negative in a climate k degrees cooler than baseline. While appreciating that unpredictable non-linearities abound in the interactions between social and climate systems (Dietz et al., 2021), I consider these simplifications much more preferable to generalized alternative models.

The gains to unbiasedness from minimally parametric models naturally come with costs in estimation efficiency; IRFs estimated by local projection regressions are fundamentally noisier than their parametric and structural counterparts such as vector autoregressions[8]. For this reason, we expect resulting estimates to be less precise than more restrictive methods, particularly for increasingly distant projection horizons since the available sample size in a fully balanced panel reduces linearly in h.

Collecting estimates of the coefficients of interest $\{\widehat{\beta}_{1h}, \widehat{\beta}_{2h}\}_{h=0}^{H}$ from (3) and adopting the notation of Jordà (2023), our set of local projection estimates im-

[7]Residuals across the projection horizon are generally autocorrelated but do not affect consistency of point estimates.

[7]Of course, under a conservative null hypothesis of non-persistence at all horizons, a permanent growth effect cannot be affirmatively evidenced with finite data and the modeler can only produce evidence of a lower bound on this persistence.

[8]Different compromises in this bias-variance tradeoff can be obtained by imposing additional restrictive parametric assumptions such as the functional approximation approach of Barnichon and Matthes (2018) which can substantially diminish estimation error.

ply a family $\mathcal{R}_{\tau \to y}(h; \overline{T})$ of state-dependent impulse responses as a function of the state variable $\overline{T}$:

$$\mathcal{R}_{\tau \to y}(h; \overline{T}) := \widehat{\beta}_{1h} + \widehat{\beta}_{2h} \cdot \overline{T} \tag{4}$$

3.1.2 Estimating state-dependent cumulative multipliers

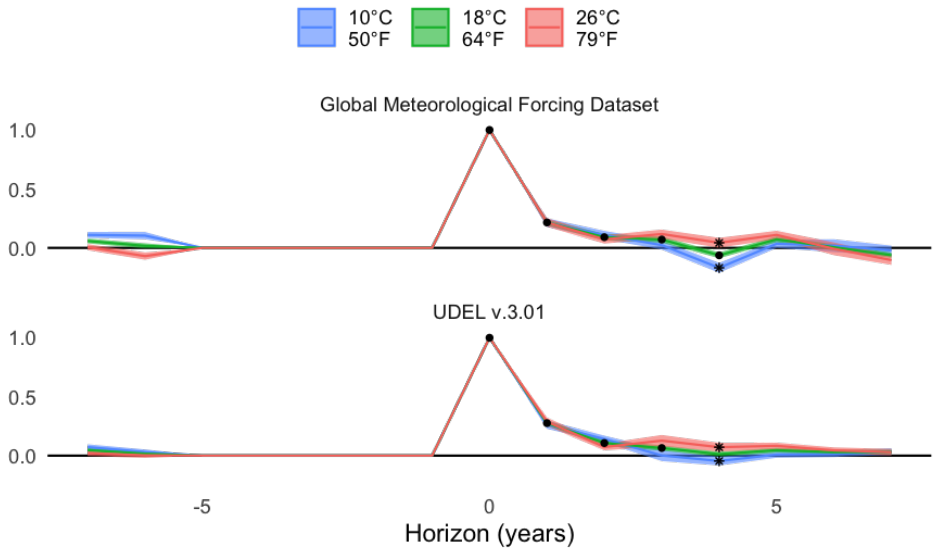
To make progress accounting for the persistence of temperature shocks themselves (see: Section 2.3), Figure 3 depicts impulse responses which result from using temperature shocks themselves as an outcome variable y as constructed by two commonly used datasets for gridded historically observed weather. We plot separate impulse responses for three representative states corresponding to the 20th and 80th percentiles of country-year expected temperatures in our sample (10°C colored blue and 26°C colored red) and their average (18°C, roughly the 40th percentile and colored green). This coloring scheme and these reference temperatures will be used throughout the rest of this paper. [9]

Points along the IRFs depicted in black correspond to point estimates which are deemed significantly persistent by a joint hypothesis test with $\alpha = 0.05$. The underlying test is based on the significance-band test recently introduced in Inoue et al. (2024) and adapted for our state-dependent setting as described in Section A2.2. In short, we have adopted a convention where filled circles correspond to significant main effects and asterisks correspond to significant interaction effects under a joint null hypothesis of non-persistence at each horizon. The presence of a significant effect up to horizon h is interpreted as evidence of a lower bound of h-period persistence. Thus, the IRFs presented in Figure 3 constitute significant evidence that temperature shocks positively influence successive temperature shocks for at least four periods although the magnitude of this persistence is mostly diminished within 1-2 years.

As emphasized by Nath et al., we will want to directly account for this persistence when attributing social impacts to a unit temperature shock. An analogy can be drawn to the estimation of multipliers in empirical macroeconomics: when estimating the impulse response of, for example, GDP to a fiscal policy shock, one must account for the tendency of stimulus programs to be implemented in stages or to be followed by additional, if typically smaller,

[9] Note that for expositional convenience, we have transformed our state variable to center on the intermediate reference temperature of 18°C so that the responses in green correspond to the sequence of estimated main effects $\widehat{\beta}_{1h}$ while the responses in blue and red incorporate equivalent but opposite interaction effects corresponding to ±8°C deviations from this reference state

Figure 3: State-dependent response of temperature shocks to a 1°C temperature shock



State-dependent impulse responses of identified temperature shocks estimated using two commonly used temperature datasets. Dots in black denote significant evidence of autocorrelation and asterisks denote significant state dependencies in this autocorrelation. Fo both products, persistence is large in magnitude for the first 1-2 periods. Negative horizons are included as a check for significant pre-shock trends.

rounds of stimulus. Otherwise, the effect sizes implied by simple impulse response functions misattribute cumulative changes in the outcome only to the initial unit shock; the calculation of multipliers scale these effects at each horizon by the corresponding accumulation of shocks. As such, we will want to present our results in the form of these cumulative temperature multipliers rather than as raw impulse response functions.

In Section A2.3, I describe how I modify methods for efficiently estimating cumulative temperature multipliers in order to implement this transformation while retaining the ability to statistically test for persistence. Ultimately, the process of estimating cumulative temperature multipliers for some outcome y amounts to estimating a cumulative variant of the original single-equation model in (3). That is, for every discrete income quantile q of interest and projection horizon $h \in \{0, 1, ..., H\}$, we estimate the following state-dependent model:

$$\Delta_h^c y_{i,t+h}^q = m_{1,h}^q \widehat{\tau}_{i,t+h}^c + m_{2,h}^q \widehat{\tau}_{i,t+h}^c \overline{T}_{it} + \lambda_h \overline{T}_{it} + \mathbf{Z}_{it}^q \boldsymbol{\gamma}_h^q + \mu_i + \eta_t + \varepsilon_{i,t+h}^q \tag{5}$$

Here, variables denoted by a superscript c are cumulative versions of the original variables constructed using the one-step process summarized by Algorithm A1 in Section A2.3. The quantile-specific matrix of controls $\mathbf{Z}_{it}^q$ include lags $\ell \in \{1, ..., L\}$ of the shocks, the interaction of the shock with the state variable, the outcome $\Delta y_{i,t-\ell}^q$, and country-level aggregate growth $\Delta y_{i,t-\ell}$.

State-dependent cumulative temperature multiplier estimates $\{\widehat{m}_{1,h} + \widehat{m}_{2,h} \overline{T}_{it}\}_{h=0}^H$ describe the effect of an identified temperature shock on the cumulative stock of outcome y^q after h periods as a function of the expected temperature.

3.2 Empirical results: evidence for a temperature effect on within-country inequality

Panel (a) of Figure 4 summarizes the results of estimating separate sets of state-dependent cumulative temperature multipliers for all 10 within-country income deciles. Here, color-coding follows the same intuitive scheme as previous plots with blue and red corresponding to representative expected temperatures of 10 and 26°C respectively[10]. The points in asterisks indicate significant interactions which impose differentiated effects based on expected climate. These effects persist for between 1-3 years, concentrating negative impacts in hot countries while keeping cooler countries relatively insulated, consistent with the aggregate results standard in the climate literature and the dynamic GDP results reported in Section A2.3. Estimation error is larger than

when estimating GDP impacts in Figure A2, unsurprising given the difference in temporal coverage and data quality.

Importantly, we are now able to compare impacts across quantiles within countries. Focusing on hot-climate countries, the estimates reported in Panel (a) imply an incidence pattern wherein negative effects of a transitory temperature shock to hot countries are found at all quantiles but are particularly dire at the bottom. For 26°C countries, the 90% confidence intervals for the impact of a temperature shock on cumulative income a year after the shock is $[-10.0, -2.9]$ percent for the bottom decile compared to $[-3.2, 0.3]$% for the second decile. This generally improves as one moves up the income distribution, stabilizing at approximately $[-2.4, 0.6]$ percent by the fourth decile through to the 99th percentile. Interestingly, the estimated impacts then discontinuously drops to $[-3.9, -0.5]$ for the richest percentile. Panel (b) shows this discontinuity within the top decile and also indicates this effect is significantly persistent for two additional years. In contrast, cumulative growth incidence is much flatter across the income distribution with point estimates suggesting positive impacts are somewhat larger for the middle classes between the third and sixth deciles.

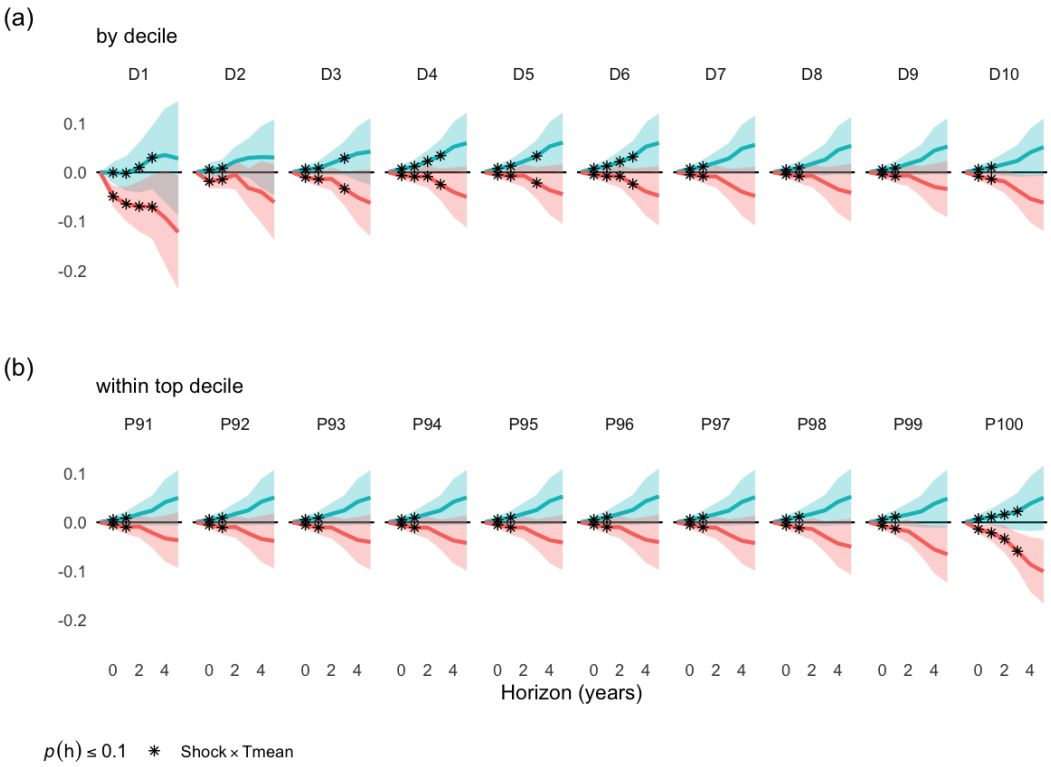
Figure 5 and the bottom panel of Figure 5 capture these discontinuities at the extremes more clearly by plotting point estimates of percentile-level impacts over time for our three representative climates. These figures imply that negative expected impacts to hot countries are indeed concentrated on both the bottom 10% and the top 1% of the distribution[11]. To the best of my knowledge, this particular vulnerability of the highest incomes in hot and generally poor countries to transitory environmental shocks has not been previously observed or theorized although it is consistent with recent work outside environmental economics demonstrating this group's disproportionate sensitivity to aggregate conditions even compared to other members of the top income decile (Alvaredo et al., 2018; Roine et al., 2009). In ongoing work, I investigate the extent to which this may be driven by weather-induced capital depreciation, capital shares in agricultural income, and the relative unavailability of weather derivatives in these settings.

This first set of results describe inequalities in the growth incidence of impacts across a national income distribution as a function of the local climate.

[10] We omit the green 18°C main effect to avoid over-busy plots, but point estimates are implied as the average of the two depicted curves due to the linearity of the interaction effect.

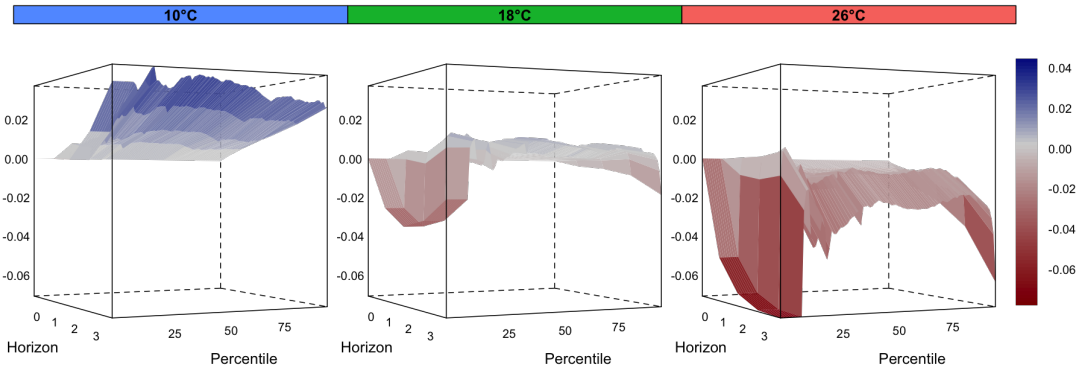
[11] As mentioned in Section 2, the income data for the bottom 10% is not well suited for decomposition into percentile effects but one would expect an ideal decomposition would reveal a similarly steep within-decile gradient in which negative impacts are driven by the poorest percentiles within the bottom 10%.

Figure 4: Temperature multipliers of national income by quantile

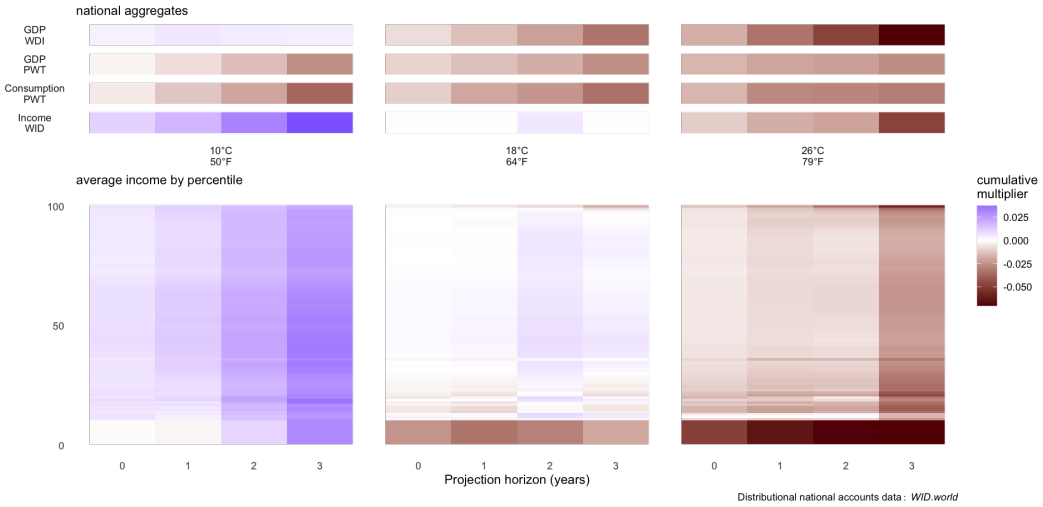


Estimates of the cumulative temperature multiplier for different quantiles of national income for two representative 'states' each: a 10°C country in blue and a 26° country in red. Asterisks denote statistically significant state-dependencies. The top row (a) depicts results for each decile of the national income distribution while the bottom row (b) depicts results decomposing for each percentile within the top decile (i.e., D10 in the top row).

Figure 5: Temperature multipliers of national income by quantile, point estimates



(a) Temperature multipliers of income. 3D surfaces represent estimates of the temperature multiplier of income for the bottom decile and percentiles 11-100 of a national income distribution. Curves in the multiplier-horizon plane represent dynamics for different income quantiles while curves in the multiplier-percentile plane are akin to h-period growth incidence curves attributable resulting from a pulse of temperature. Columns correspond to different representative climates.



(b) Aggregate vs. distributional multipliers. Planes at the bottom flatten the surfaces in panel (a) by using color gradients to represent positive (blue) and negative (brown) multipliers. Strips at the top are the equivalent multipliers estimated for country-level economic aggregates.

The disproportionate impacts to the bottom decile strongly imply a regressive effect: In the aftermath of an isolated temperature shock, a representative member of the the bottom decile in a 26°C country can expect to earn 7% less in total over the next three years than they would had the shock never occurred. This compares to losses of 2.2% for the median adult in their country and a *gain* of 3.0% for their counterpart in a country with a 10°C climate. At the same time, a representative member of the 100th percentile in the same 26°C country, represented by the thinnest sliver at the top of the rightmost distributional contour plot in (b), can expect to earn 5.9% less over those three years than they would otherwise compared to 3.6% for the 99th percentile.

4. Counterfactual analysis

4.1 From identified temperature shocks to systematic climate change

The results presented in Section 3.2 are expressed as responses to a 1°C shock, which notably is much larger in magnitude than shocks we would expect to observe in a typical year in any country. Referring back to the time series of shocks illustrated in Figure 1, the median magnitude of a shock varies by the rate of climate change which will in turn vary across geographies and time. For the warmest 33% of country-years in our data, the median shock was approximately 0.19°C in absolute value in the 1970s and then increased to between 0.30-0.37° in the ensuing decades. Thus, hot-country effects can be understood as responses to a shock 3-5 times larger than is typical. For the coldest 33% of country-years, the median magnitude of a shock was approximately 0.35°C in the 1970s and 1980s, 0.51-0.53°C in the 1990s and 2000s, and 0.59°C in the 2010s. Their effect sizes then can be interpreted as the response to a shock just 2-3 times larger in magnitude than the median.

So while those results consistently demonstrated much larger distributional responses to the same shock for residents in warmer geographies, this disparity is at least partially offset by the relatively mild incidence of shocks in these settings. The counterfactual analysis we document in this section is valuable as a means of interpreting these reduced-form results practically, accounting for the spatial incidence of temperature shocks as well as other factors such as the absolute distribution of global incomes across country-quantiles and the spatial distribution of the global population across different geographies as observed in the historical record.

In the absence of a structural climate-economy model, it is standard prac-

tice to extrapolate reduced-form effects estimated using short-term weather variation to infer long-term impacts of sustained climate change. Broadly, this involves applying the estimated impact model to a counterfactual climate scenario to produce a counterfactual trajectory for the social outcome of interest. The difference between the social outcome of interest under this counterfactual relative to its trajectory under a reference scenario (e.g., for historical attribution studies like the present analysis, the actual historically observed trajectory of this outcome) are then attributed to the differences in the climate under the two scenarios.

In our case, we are estimating the effect of changes to the global climate system between 1980 and 2016 on the global distribution of income. Using the observed history of country-year temperatures, we calculate the implied history of country-year termperature shocks as defined in (1a) and (1b). For the bottom national income decile and each of the 11th to 100th national income percentiles, we collect $B = 500$ bootstrap estimates of the state-dependent cumulative temperature multipliers of income according to (5). Integrating these percentile-bootstrap specific estimates to the observed histories of temperature shocks implies a counterfactual history of country-percentile-year income absent any temperature shocks between 1980 and 2016. The differences between this simulated history of global income distributions and the observed history of global income distributions are then attributed to changes in the climate over this 37-year period.

This process (and its analogous implementation for projecting future climate change impacts) offers several significant advantages over prior approaches to counterfactual attribution. First, since estimates of cumulative damages result from local projection regressions, dynamic impacts are identified directly from medium-term long differences in income and disciplined by a statistical test of their persistence rather than imposed by modeling choices as described in Section 2.4.

Second, models which use levels of temperature as their treatment variable have been challenged on the grounds that these counterfactual exercises require them to extrapolate effects of treatments increasingly beyond the range of observed data. This is much less of a concern in our setting where the treatment variable is an identified temperature shock which is dynamically anchored by a time-varying state variable. As a result, the treatment variable remains largely within the range of the historical record even under rapid warming because the state variable gradually adjusts to recent experience. In this sense, our model accommodates a notion of adaptation in the sense of S. M. Hsiang (2016) wherein the "direct" effects of exposure to a temperature shock can be partially offset by Bayesian "belief effects"; the same tempera-

ture level is less of a 'shock' the more experience one has with similar temperatures. In our case, a one-time discrete shift from one climate to a new climate would be fully internalized after 30 years.

More rapid adaptation can be accommodated by lowering the value of M defining the state variable. Alternatively, Bilal and Känzig (2024) and Nath et al. (2024) define temperature shocks as the residuals from autoregressive distributed lag models (ARDLs) of temperature levels. In so doing, they impose a model of much more immediate and complete adaptation so that if a $26°$C country were to be exposed to, for example, $40°$C weather five years in a row, the fifth year will impose no economic cost because it would register as a $0°$C shock. This difference in assumed adaptability is why their time series of historically observed temperature shocks is largely homoskedastic around a constant mean of approximately zero while ours is positively trended in periods of warming (as seen in Figure 1).

Our more conservative construction offers a number of attractive advantages. First, it is arguably more consistent with the empirical literature on climate adaptation; a recent review of this work finds "limited evidence" of observable adaptation moderating the direct impacts of weather shocks in the context of climate change (Burke & Emerick, 2016). Secondly, it lends itself to a simple, intuitive, and consistent approach to estimating damages from climate change: the empirical analysis isolates an identified temperature shock to estimate its social impacts and then the counterfactual analysis represents climate change as a positive trended bias in the distribution of these shocks and uses the empirical result to calculate damages.

This intuition is not available to approaches which define temperature shocks as residuals from an ARDL. Because the time series of temperature shocks reverts to a mean of zero within the periodicity of its maximal lag, the distribution of temperature shocks are nearly homoskedastic and symmetrical about 0 even amid climate change. As a result, the same intuitive approach to counterfactual analysis would imply near-zero cumulative social costs from climate change.[12]

[12]To arrive at non-zero climate damages, Bilal and Känzig (2024) and Nath et al. (2024) instead seem to use the conventional "delta method" appropriate for parametric models which use temperature levels as their treatment. This entails first linearizing the difference in temperature levels before and after the relevant period of warming; for example, under the most extreme warming scenario between 2010 and 2100, the population-weighted mean temperature increase over this period is $4.3°$C. These differences are then seemingly interpreted as a representative long-term 'shock' to be multiplied by the coefficient on the original temperature shock.

4.2 Results

In constructing these counterfactual series, each bootstrap b assigns each country-quantile-year to a global percentile corresponding to their population-weighted income ranking across the global income distribution for that year. Finally, we calculate the long-term growth between 1981 and 2016 implied by each series for each bootstrap series. The resulting long-term growth incidence curve is depicted in Figure 6.

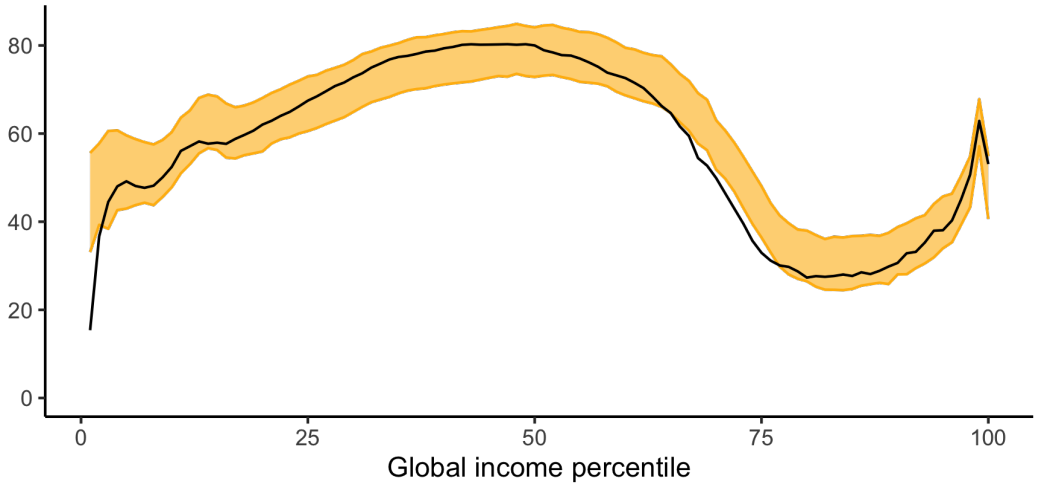
These results may be interpreted as impacts of observed climate change insofar as it the influence of climate change is sufficiently well-captured by asymmetries in the incidence of temperature shocks and changes to expected temperatures over the 36-year period of analysis. Section A4 presents additional results decomposing these effects into their anthropogenic and natural contributions by applying methods from climate attribution science to output from state-of-the-art climate models, arriving at largely the same conclusions.

Here, the historically observed level of growth over this period is represented by a black filled line, reproducing a version of the "elephant curve" famously documented in Lakner and Milanovic (2016). The blue band depict the 90% confidence interval for the counterfactual removing the influence of temperature shocks on income. Regions where this band is above the observed incidence curve imply deprivations of income attributable to the incidence of temperature shocks over this period. Point estimates imply probable net harm for around 65% of the global sample.

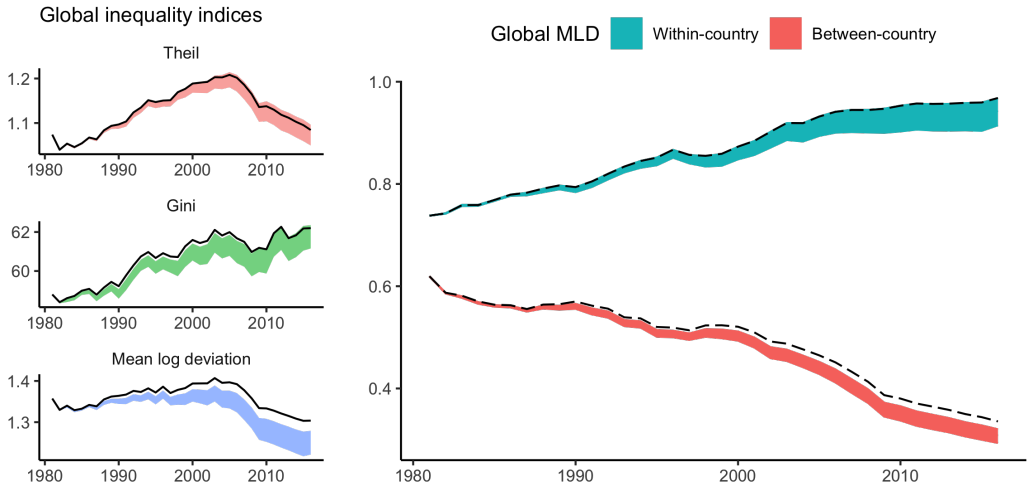
Despite proportionally milder absolute warming in the poorest countries (see Figure A5), this negative incidence still concentrates on the world's poorest 20%, representing 1.2 billion people in our 2016 sample and mostly located in warm developing countries. Importantly, as in Alvaredo et al. (2018), calculations underpinning this figure omit the bottom 10% of each country's income distribution because of the inclusion of unemployed adults and compositional instability effects which arise from assigning deciles of very large populations to percentiles representing smaller populations. For the poorest percentile in our data, we find that incomes absent these disproportionate temperature shocks would be 29% $[18, 41]$ higher than they are today. As these effects exclude precisely the subpopulation we have already found to be the most economically vulnerable to temperature shocks, this growth incidence curve should be interpreted as a likely very conservative estimate of deprivations to the global poor.

Elsewhere, estimates also imply likely harm from warming for the 52nd-97th percentiles. That the top 3% are relatively unaffected despite the negative effects for the top 1% in Section 3 is because these groups are dominated by ex-

Figure 6: Global income inequality (1980-2016), observed vs. zero-warming counterfactual



(a) Change in income levels (%), 1980 vs. 2016. Black line depicts observed increases in income levels by global income percentile among positive income-earners. Bands in gold depict the 95% confidence interval of the counterfactual increase had warming stayed fixed at 1980 levels.



(b) Global inequality, total and decomposed. On left, black lines depict the observed evolution of global inequality from 1980 to 2016 by three common inequality indices. Colored bands represent the 95% confidence interval for their evolution under the same counterfactual. On right, we decompose the mean log deviation series into its between-country and within-country contributions.

treme incomes in relatively cold countries. Effects on the global middle class are relatively tempered with slightly positive point estimates estimated for essentially the entire second quartile, also consistent with the within-country incidence curves which found the middle class experienced the mildest economic impacts across all climates.

Despite these different-signed effects, the counterfactual growth incidence band is noticeably flatter than the observed elephant curve, implying that historical climate change has been globally regressive in addition to acting as a drag on global economic growth. Figure 6 plots the time series for the global equivalents of the within-country inequality measures introduced earlier. In our within-country analysis, we identified regressive transfers from the bottom decile and progressive transfers from the top percentile. The historical effects represented in these graphs summarize how these distributional patterns aggregate globally according to index-specific social welfare functions[13]. All three measures are increasing in inequality and for each, redistributive effects are found to be large and regressive: proportionally, the global Theil index increases by 0.9% $[-1.8, 2.8]$, the Gini index by 0.5 points $[-0.2, 1.0]$, and the mean logarithmic deviation index by 3.9% $[1.6, 6.4]$.

The right panel in Figure 6 decomposes the effect on the global MLD index into its within-country and between-country dimensions. We find that climate change has unambiguously exacerbated both varieties of inequality, increasing between-country inequality by 8.7% [4.9, 13.3] and within-country inequality by 2.6% [0.0, 5.6]. To the best of my knowledge, this represents the first empirical estimate of the latter quantity in the vast empirical literature on climate impacts.

[13] One interpretation of the Theil measure is as a geometric measure of inequality. For any given income distribution, an individual with $100,000 giving $10 to someone with $10,000 would improve the Theil inequality index by the exact same amount as someone with $1,000 giving the same $10 to someone with $100 because the ratio of incomes between the donor and recipient are the same in both cases (Cowell, 2011).

Conversely, a $100 income earner regressively transferring 10% of their income to a $1,000 income earner would offset the progressivity of a $100,000 income-earner transferring a ten-thousandth of their income to a $10,000 income earner.

[13] One interpretation of the mean logarithmic deviation measure is as the average disutility of a given income distribution compared to an equal distribution of the same total income, assuming logarithmic individual utility. Of the three measures, the Theil places the most welfare weight on redistribution among the top quantiles while the MLD places the most welfare weight on redistribution among the bottom quantiles.

5. Conclusion

The results presented in this paper constitute the first comprehensive evidence of regressivity in the global incidence of climate change, complementing an expansive and active economic literature which has remained effectively agnostic on global distributional impacts at subnational levels. By importing methods recently developed in empirical macroeconomics to distributional data thus far largely unused in environmental economics, we were able to overcome several long-standing statistical and practical constraints which have long obstructed progress on this urgent research question.

Given the sparsity of other work on this broad topic, several limitations of the present study suggest critical areas for future research. First, while our evidence affirms and sharpens the conventional understanding that the world's poorest have been and will continue to be by far the most severely impacted by climate change, our analysis remains unsatisfyingly imprecise about these effects. Despite the impressive extent of the new World Inequality Database, addressing the inability to distinguish between those reporting no income and overcoming the limitations of even ideal income data as a proxy for welfare among the global poor are outstanding priorities in the climate impacts literature.

In revisions in progress, the distributional analysis reported here will be supplemented by an analysis of compositional effects identifying which income groups from which countries have been transitioning up and down the global income distribution as a result of climate change. If income effects are as severe and state-dependent as described here, then it becomes plausible that an important distributional consequence of climate change is that inhospitable environments becomes increasingly deterministic of economic development, severely limiting the upward mobility of populations in hotter climates.

To the best of my knowledge, the vulnerability of the highest incomes in hot and generally poor countries to transitory environmental shocks has not been previously documented or theorized. Weather-induced capital depreciation, capital shares in agricultural income, and the relative unavailability of insurance instruments such as weather derivatives in developing countries are example plausible mechanisms that call for empirical testing.

From a policy perspective, the results presented in this paper suggest that incorporating equity considerations as through a concave utility function would indeed increase the social weighting of the most impacted by climate change. But it is may still be worth taking seriously criticisms that a "profound indifference to inequality" (Sen et al., 2020) remains embedded even in these

progressive varieties of cost-benefit analyses predominant in the climate economic literature. For example, Prest et al. (2024) show that one proposed implementation would increase the US government's official estimate of the social cost of carbon by a factor of eight. But somewhat paradoxically, these greater cost estimates are primarily used to prescribe an increased price on carbon which may then disproportionately burden these same segments of the population whose increased welfare weight induced the inequality-exacerbating policy. Equity weighting should of course factor into both the estimation of costs and the design of policy internalizing them.

To this end, it is intuitive that democratic pressures and a richer portfolio of policy instruments should be more effectively mobilized domestically than coordinated across a Westphalian international system (Barrett, 2003; Keohane & Victor, 2011; Levin et al., 2009). If so, levers available domestically but not internationally—such as compensatory fiscal transfers—may yet emerge as viable instruments of climate policy.

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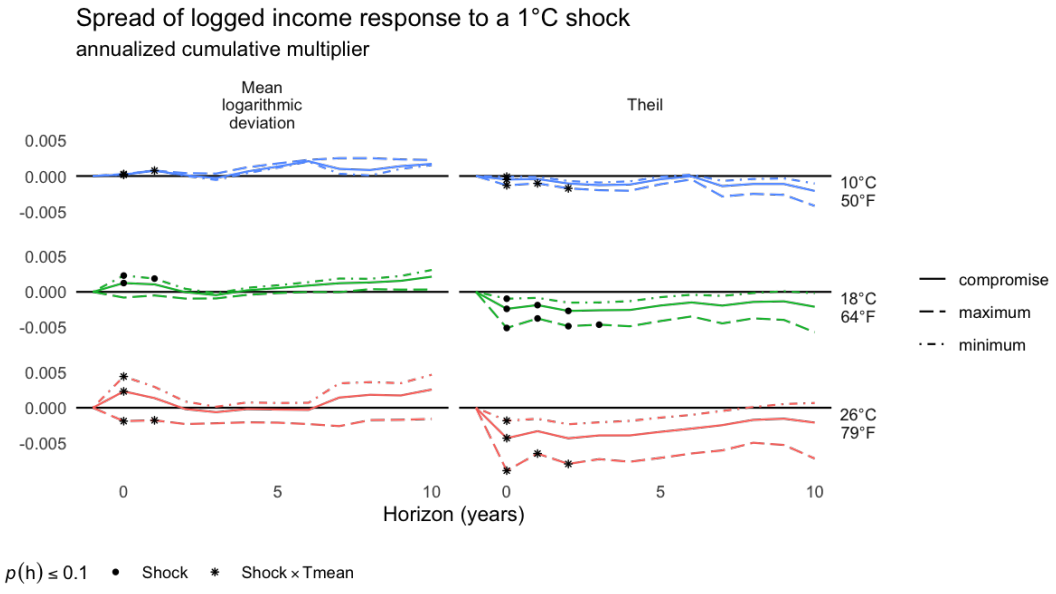
A1. Data

A1.1 Construction of inequality indices

Data from the World Inequality Database can be extracted at the country-percentile level with variables reporting averages, minimums, maximums, and shares of national income. As grouped data, its use for the construction of inequality indices still requires the modeler to make assumptions about the distribution of household incomes within these disaggregated groups. This becomes increasingly influential for the construction of entropy-based inequality measures such as the mean logarithmic deviation or the Theil index for higher levels of inequality across groups.

For a given inequality measure J, the lower bound J_L is calculated by assuming perfect equality within all groups such that all individuals are assigned their group mean income. The upper bound J_U is calculated by assigning a proportion of the individuals within a group the minimum income of the group and the remainder the maximum income of the group.

Figure A1: Sensitivity of inequality indices to within-quantile inequality



Cowell, 2011 describes multiple 'compromise' assumptions between the

two extremes, finding that they each approximate an average where the lower bound is given twice the weight of the upper bound. Figure A1 depicts IRFs of the two inequality measures under each of these three assumptions. For our purposes, we use the compromise construction whenever a choice is necessary.

A2. Methods

Here, we provide additional exposition of our application of the method of local projections for inferring a dynamic social response to the temperature shock described in Section 2.3. The supplementary analysis here uses country-level GDP from the Penn World Tables (Feenstra et al., 2015) as the outcome variable of interest. The results documented here may also be compared to other estimates in the prolific literature on global climate impacts estimated by top-down methods, which most commonly use similar country-level aggregates as outcomes.

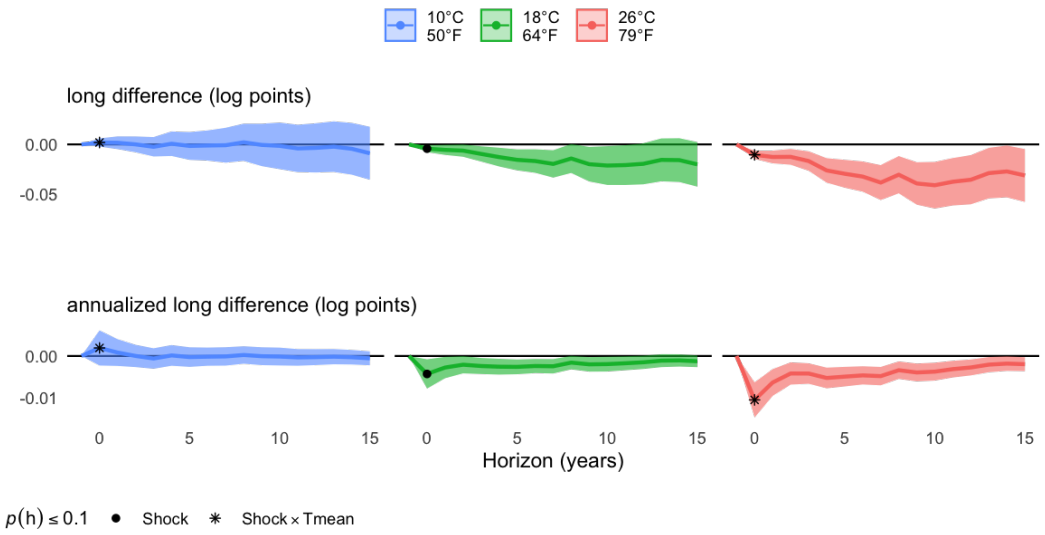
A2.1 State-dependent impulse response of GDP

We first estimate (3) on a country-year panel dataset using the h-period long difference in the logarithm of GDP as our outcome variable $\Delta_h y_{i,t+h}$. Figure A2 illustrates the estimated impulse response of GDP up to a projection horizon of 6 years after a unit temperature shock. Colors correspond to three representative temperatures depicted in different colors. The plots in the second row depict the same functions annualized by simplying dividing estimates at each projection horizon h by $h + 1$.

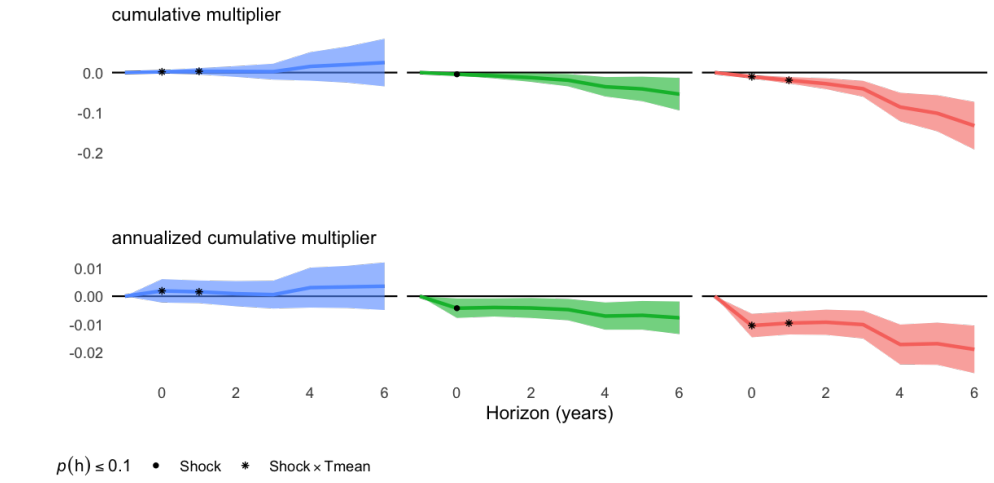
The contemporaneous estimates in the IRFs in the first row of Figure A2 depict a pattern now standard in the aggregate impacts literature featuring near-null temperature effects in cold climates, mild effects in mid-temperature climates, and the largest negative effects in hot climates. The evolution of these impacts over time is less commonly depicted and here produce a pattern qualitatively similar to those reported by Nath et al. wherein per capita GDP appears to stay persistently depressed over the entire projection horizon. The second row depicts the same estimates but annualized by scaling down estimates by their corresponding horizon $h + 1$ to more clearly depict convergence behavior relative to the contemporaneous effect of the shock.

Global inequality effects of climate change thus are not just dependent on the shape of the impulse response responses across quantiles but also the distribution of warming over space and time as well as the actual distributions

Figure A2: State-dependent dynamic responses to a 1°C shock



(a) State-dependent impulse response of GDP. The plots in the top row correspond to estimates of impulse responses of GDP estimated by the $H+1$ local projections regressions specified in (3) for three representative temperatures depicted in different colors. The plots in the bottom row depict the same estimates and error bands annualized by dividing values at each horizon h by $h+1$



(b) State-dependent temperature multiplier of GDP. The cumulative multiplier is interpreted as the change in *cumulative* GDP over h periods attributable to a unit shock in $\widehat{\tau}$, net of its impact on future shocks $\widehat{\tau}$. Annualized multipliers represent the same estimates but divided by the projection horizon.

of incomes across quantiles. The simulation exercises to be described in Appendix 4 are valuable because they account for four relevant climate inequalities: inequality of temperature effects across an income distribution, inequality in the incidence of temperature effects across different climates, inequality in the global distribution of income, and spatial inequality in the incidence of absolute warming.

A2.2 Inferring persistence from local projection estimates

The colored error bands depicted in Figure A2 reflect pointwise estimation error. This is particularly important to include in this context since the volume of regressions underlying a single LP-IRF precludes convenient summary in a single regression table.[14] One may be tempted to infer from the fact that the hot-country bands in red do not intersect the horizontal axis over the entire projection horizon that these estimates constitute significant evidence to reject a joint null hypothesis of level effects which persist for fewer than 15 years.

However, the uncertainty depicted by error bands corresponds only to individual hypothesis tests. Since estimates and residuals across an IRF are serially correlated—for example, the green bands straddle the horizontal axis but all estimates are negative— the relevant test of significance is a joint hypothesis test not represented by pointwise error. The significance of the main effect of the impulse response $\mathcal{R}_{\tau \to y}(h)$ amounts to testing the null hypothesis that all main effects up to h are 0:

$$H_0(h) : \beta_{1,0} = \beta_{1,1} = ... = \beta_{1,h} = 0$$

Since the number of coefficients being considered in the joint hypothesis increases in h, the coverage of the joint hypothesis test must be adjusted for the inclusion of each additional horizon to compensate for the problem of multiple hypothesis testing. Inverting this joint null hypothesis, Inoue et al. (2024) derives convenient estimators for the implied "significance band" defined to satisfy the following condition:

$$\mathbb{P}\left[\bigcap_{h=0}^{H}\left\{\zeta_{\frac{\alpha}{2(H+1)}}\frac{\sigma_h}{\sqrt{T-h}} < \widehat{\beta}_h < \zeta_{1-\frac{\alpha}{2(H+1)}}\frac{\sigma_h}{\sqrt{T-h}}\right\}\right] \geq 1-\alpha$$

[14] For example, the results depicted in Figure A2 derive from 16 different regressions. Later results derive from as many as 91. All graphically depicted errors and significance tests presented in this paper correspond to a significance level of $\alpha = 0.1$.

The scaling of the significance level α by $2(H+1)$ is an implementation of the Bonferroni correction. In practice, rejecting the null for the main effect corresponding to horizon $h+1$ is sufficient to reject the null of non-persistence up to horizon h. Graphically, these bands are represented by a region straddling the horizontal axis such that estimates of β_{1h} contained within it are insufficiently large enough to reject the hypothesis of a null effect that the impulse response persists up to period h, analogous to the bands drawn to measure persistence in time series correlograms. The width of these bands increases with the projection horizon because the available estimation sample for a near-balanced panel is reduced by one period per unit for each additional projection.

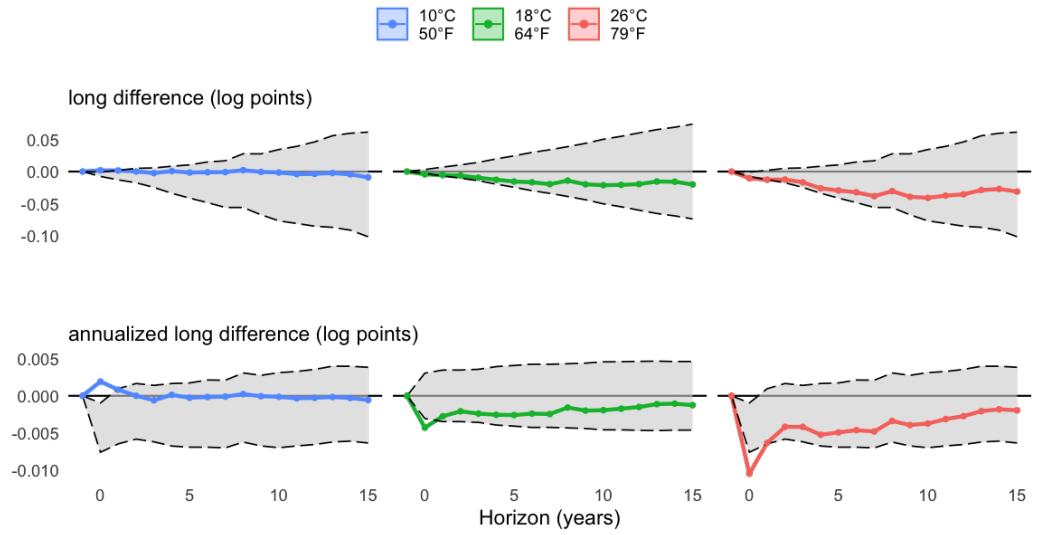
These significance bands are specific to the main effect and are represented by the gray regions in the middle panels of Figure A3. Since ours is a state-dependent model, I derive equivalent significance bands specific to the interaction effects. These are defined relative to the point estimates of the main effects and are depicted by the gray regions in the panels on the left and right. Points outside these regions are considered sufficient to reject the null hypothesis that interaction effects do not persist up to h periods. By the linearity assumption, the null hypothesis of zero interaction effects for a 10° climate is rejected if and only if it is rejected for a 26° climate.

For comparison, Figure A3 depicts a placebo test where I construct significance bands equivalent to those described above to test the persistence in the effect of temperature on another widely available and potentially non-stationary growth rate: that of population. As I am not aware of any theory or evidence to suggest temperature shocks would meaningfully impact country-level population growth[15], we should expect the tests to return precise null results. Indeed, we find with all point estimates resulting from two separate population datasets easily contained by the corresponding significance bands. Notably, if we were to misuse the error bands as indicators of statistical significance, we would misinterpret the estimated IRFs as evidence to reject the null for large subintervals of the projection horizon for mid-temperature and hot climates. Instead, the entirety of these error bands are easily enveloped by the significance bands.

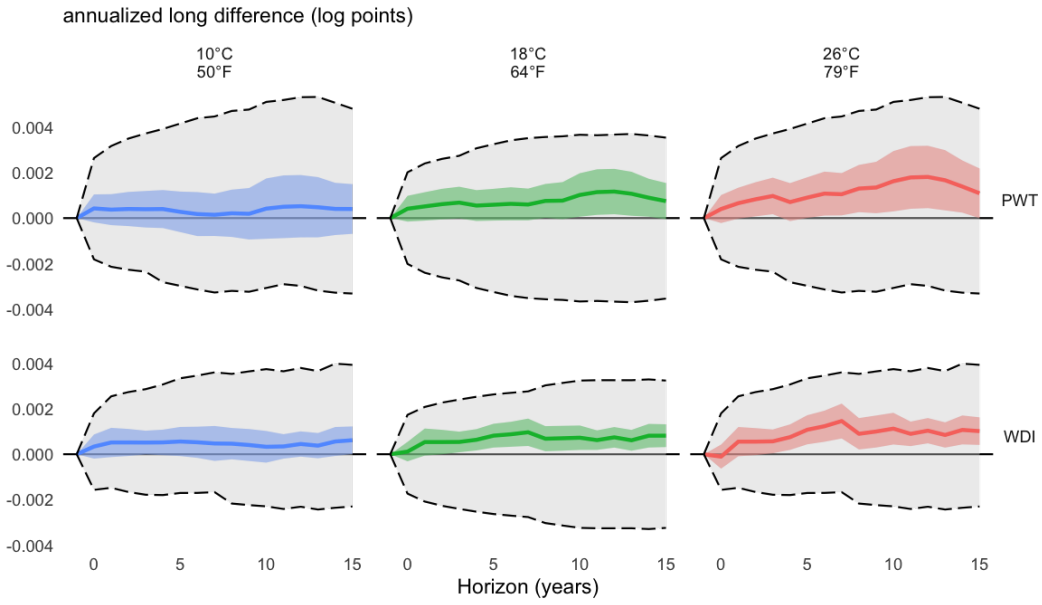
Since including significance bands, color-coded error bands, and point estimates for multiple IRFs can clutter space, I adopt a convention of omitting the significance bands entirely and instead coloring black the point estimates which are located outside the implicit error bands. In all IRFs beginning with

[15]While there is an established literature on climate impacts on mortality rates (A. Barreca et al., 2016; A. I. Barreca & Shimshack, 2012; T. Carleton et al., 2022), absolute magnitudes are miniscule relative to total populations

Figure A3: Impulse responses to a 1°C shock, including significance bands



(a) GDP impulse response. Gray significance bands in the middle panels correspond to the threshold magnitudes for significant main effects while those in the left and right panels are centered on the point estimates of these main effects (green filed lines) and correspond to the threshold for significant state interaction effects.



(b) Placebo test: national population response. Equivalent to the bottom row of panel (a) but for national population, an outcome variable we do not expect temperature to affect and with conventional error bands also included. Rows differ by source of population data.

those depicted in Figure A2, rounded points correspond to significant main effects while asterisks correspond to significant interaction effects. By this test, these results cannot reject the null hypothesis of one-period level effects.

A2.3 The state-dependent temperature multiplier

Another reason to prefer temperature shocks to levels of temperature as treatment variables is that models which use the latter neglect to sufficiently account for autocorrelation in temperature itself when estimating dynamic effects (Bilal & Känzig, 2024; Nath et al., 2024). An analogy can be drawn to the estimation of multipliers in empirical macroeconomics: when estimating the impulse response of, for example, GDP to a fiscal policy shock, one must account for the tendency of stimulus programs to be implemented in stages or to be followed by additional stimulus. Otherwise, the effect sizes implied by simple impulse response functions misattribute cumulative changes in the outcome only to the initial unit shock; multipliers scale these effects at each horizon by the corresponding accumulation of shocks.

Figure 3 depicts the impulse response of temperature shocks to a pulse of itself for our three representative climates. Each exhibits very similar persistence patterns across two independent temperature dataset. An application of the persistence test covered in Section A2.2 implies autocorrelations which persist for up to 10 years though only the first 1-2 periods are qualitatively large (approximately 21% and 9% respectively with insignificant heterogeneity across climates).[16] Cumulative multipliers are estimated in order to scale down the estimated impulse response of GDP $\mathcal{R}_{\tau\to y}(h)$ by the impulse response of the shock to itself $\mathcal{R}_{\tau\to\tau}(h)$ to account for this dynamic treatment schedule.

Cumulative responses are traditionally computed by first constructing *cumulative* versions of the two IRFs: $\mathcal{R}^c_{\tau\to y}(h) := \sum_{j=0}^{h} \mathcal{R}_{\tau\to y}(j)$ for the outcome of interest y and $\mathcal{R}^c_{\tau\to y}(h) := \sum_{j=0}^{h} \mathcal{R}_{\tau\to\tau}(j)$ for the shock $\widehat{\tau}$. Then the cumulative temperature multiplier $m_{\tau\to y}(h)$ would be calculated as the ratio

$$m_{\tau\to y}(h) := \frac{\mathcal{R}^c_{\tau\to y}(h)}{\mathcal{R}^c_{\tau\to\tau}(h)}$$

However, this approach has a couple of inconvenient downsides. First, as

[16] This contrasts with the finding of NRK who measure shock persistence using levels of temperature to infer that temperature shocks remain 10-20% higher for at least nine years following a unit shock.

noted by Jordà (2024), computation of standard errors of a ratio of two random variables is complicated and non-standard. Second, it is computationally inefficient in the sense of requiring $2(h+1)$ (or $4(h+1)$ in our state-dependent case) separate local projections to obtain the corresponding multiplier for each projection horizon h.

Instead, we follow the recommendation to use the one-step method described in Ramey (2016) and applied in Ramey and Zubairy (2018). We adapt the method for our state-dependent setting through the procedure summarized in Algorithm A1.

We can use the resulting coefficients to trace a cumulative multiplier function $\mathcal{R}^c_{\tau\to y}(h;\overline{T})$ analogous to (4) but which accounts for the effect of persistence in the shock. Since estimation of the coefficients follows directly from a system of local projection regressions, standard inference is preserved and the significance testing methods outlined in Section A2.2 conveniently remain available. Figure A2 demonstrates an application of the method to estimate temperature multipliers of GDP. The estimated multiplier over horizon h is interpreted as the difference in cumulative GDP over the h periods following an isolated 1°C shock relative to a counterfactual where the shock had never occurred.

Estimated cumulative multipliers are found to be substantial at larger horizons but the significance band test cannot reject a null of non-persistence beyond one period. 90% confidence intervals for the effect of an identified temperature shock on cumulative GDP are $[-0.49, 1.1]$ percent for 10°C climates and and $[-2.7, -1.1]$ percent for 26°C climates.

Algorithm A1: One-step state-dependent cumulative multiplier

for $h = 0$ **to** H **do**

Estimate the IRF $\mathcal{R}_{\tau\to\tau}(h)$ as defined in (4), collecting state-dependent shock coefficients $\widehat{\alpha}_{1,h}, \widehat{\alpha}_{2,h}$;

Define $\Delta^c_h y_{i,t+h} := \sum_{j=0}^{h} \Delta_j y_{i,t+j}$;

Define $\widehat{\tau}^c_{i,t+h} := \widehat{\tau}_{it}\left[\sum_{j=0}^{h} \widehat{\alpha}_{1,j} + \widehat{\alpha}_{2,j}\overline{T}_{it}\right]$;

Estimate local projection:

$\Delta^c_h y_{i,t+h} = m_{1,h}\widehat{\tau}^c_{i,t+h} + m_{2,h}\widehat{\tau}^c_{i,t+h}\overline{T}_{it} + \lambda_h \overline{T}_{it} + \mathbf{Z}_{it}\boldsymbol{\gamma}_h + \mu_i + \eta_t + \varepsilon_{i,t+h}$;

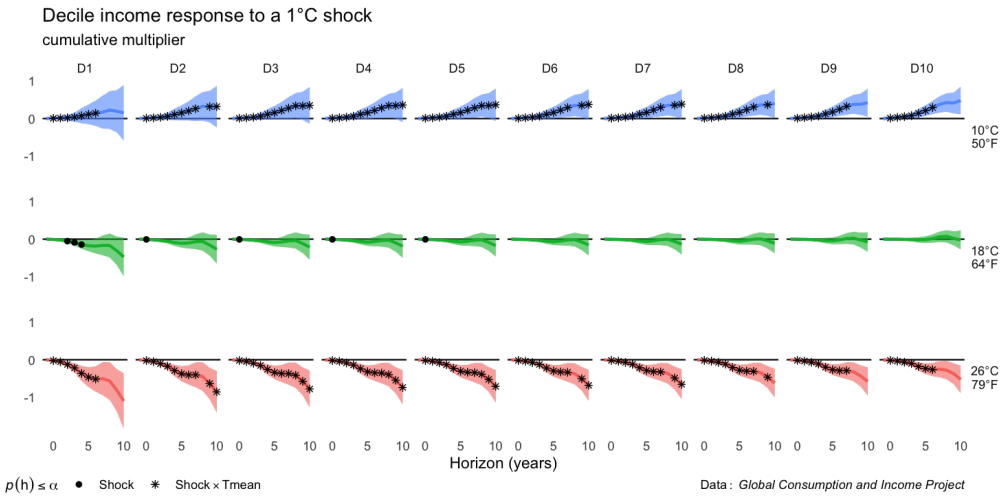
end

Output: Multiplier coefficients $\widehat{m}_{1,h}, \widehat{m}_{2,h}, \widehat{\lambda}_h$ for $h \in \{0, ...H\}$

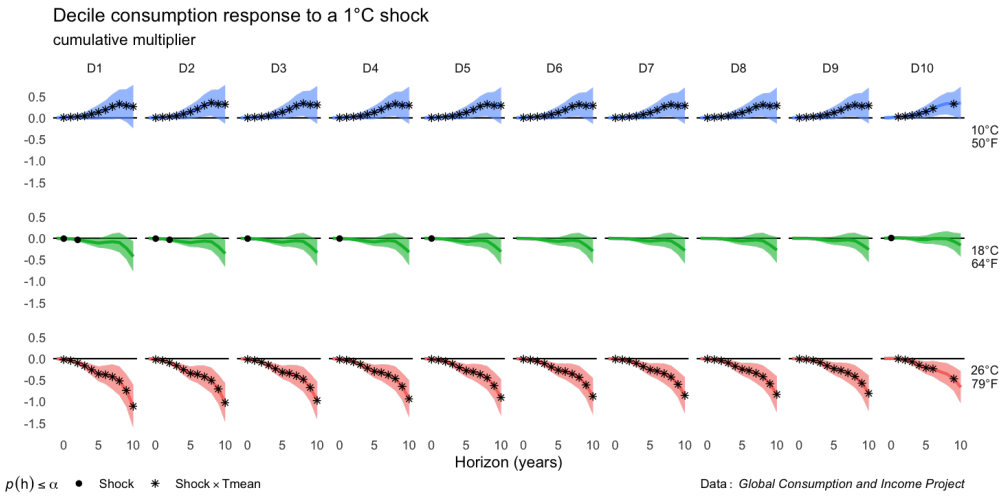
A3. Supplementary results

Figure A4: Temperature multipliers estimated using survey microdata (Lahoti et al., 2016)

(a) Impacts on income by decile



(b) Impacts on consumption by decile



A4. Anthropogenic attribution

To attribute a global inequality effect to historical anthropogenic climate change, we implement a counterfactual analysis which decomposes observed temperature effects into their natural and anthropogenic contributions, adapting methods commonly used in climate attribution science.

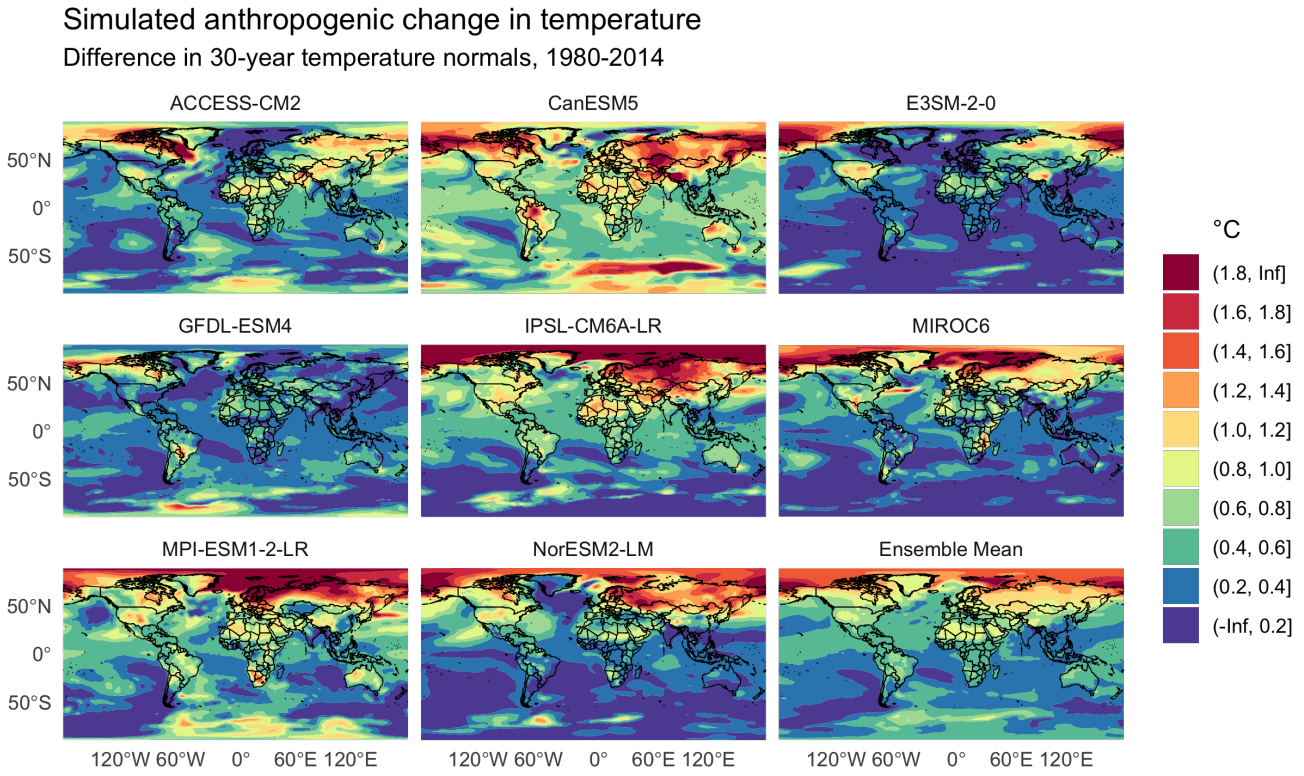
A4.1 Climate simulation data

Climate simulation data comes from the contributions of climate modeling groups to the World Climate Research Program's Coupled Model Intercomparison Project (CMIP). The CMIP, initiated in 1995, is a framework to coordinate, standardize, and disseminate the results of state-of-the-art simulations of the global climate for the benefit of the international climate research community. Data used in this study comes from CMIP6, the sixth and newest[17] generation of models (Eyring et al., 2016). Data is provided as a collection of globally gridded time series summarizing the joint evolution of several hundred climate variables under various pre-specified calibrations. We will primarily be interested in the models' simulations of near-surface air temperature.

All participating models are "coupled", meaning that they explicitly account for interactions and feedback between distinct components of the global climate system such as the atmosphere, the cryosphere, land surface, and the ocean. Because the manner in which this integration is achieved is idiosyncratic to each model, computationally expensive, and highly sensitive to external calibrations, coordinating the set of "experiments" that participating models run is necessary for comparability and interpretability. For example, one of the four indexes which define an experiment is the realization index which corresponds to the set of geophysical initial conditions at the beginning of the simulation period. By holding the initial conditions fixed, differences in results across models within the same experiment can be more readily attributed to the distinguishing features of the model without being confounded by differences in implementation. Similarly, differences in the same climate model's results across different perturbations to the initial conditions provide a measure of the model's internal variability. The CMIP thus provides a systematic way of decomposing total variation across models and experiments both internal and external to the model. Coordination also enables identification of systematic and idiosyncratic biases in model designs that then inform the next generation of models.

[17] After a delayed rollout, outputs from most CMIP6 models were made accessible by 2022

Figure A5: Model-implied spatial distributions of global warming, 1980-2014



Simulated spatial distribution of anthropogenic warming for eight randomly selected climate models. The bottom-right plots the average distribution for all 13 CMIP6 models which ran both historical and historical-natural simulations.

Among the other three indexes which define an experiment, the forcing index is central to our counterfactual analysis. Forcings are factors which affect the net transfer of radiative energy in the climate system, essentially the energy the Earth receives from the sun minus that which it expels back into outer space. Anthropogenic forcings include, for example, industrial greenhouse gas emissions, deforestation, and the transformation of land for arable agriculture. Natural forcings include aerosol expulsions from major volcanic activity and variation in solar irradiance such as those associated with the 11-year solar cycle.[18] Comparing simulations which only differ in their forcings allow one to attribute differences in outcomes to the differences in forcings; we describe next how comparisons of this kind form the basis of both our historical and projection counterfactual analyses.

A4.2 Construction of historical counterfactuals

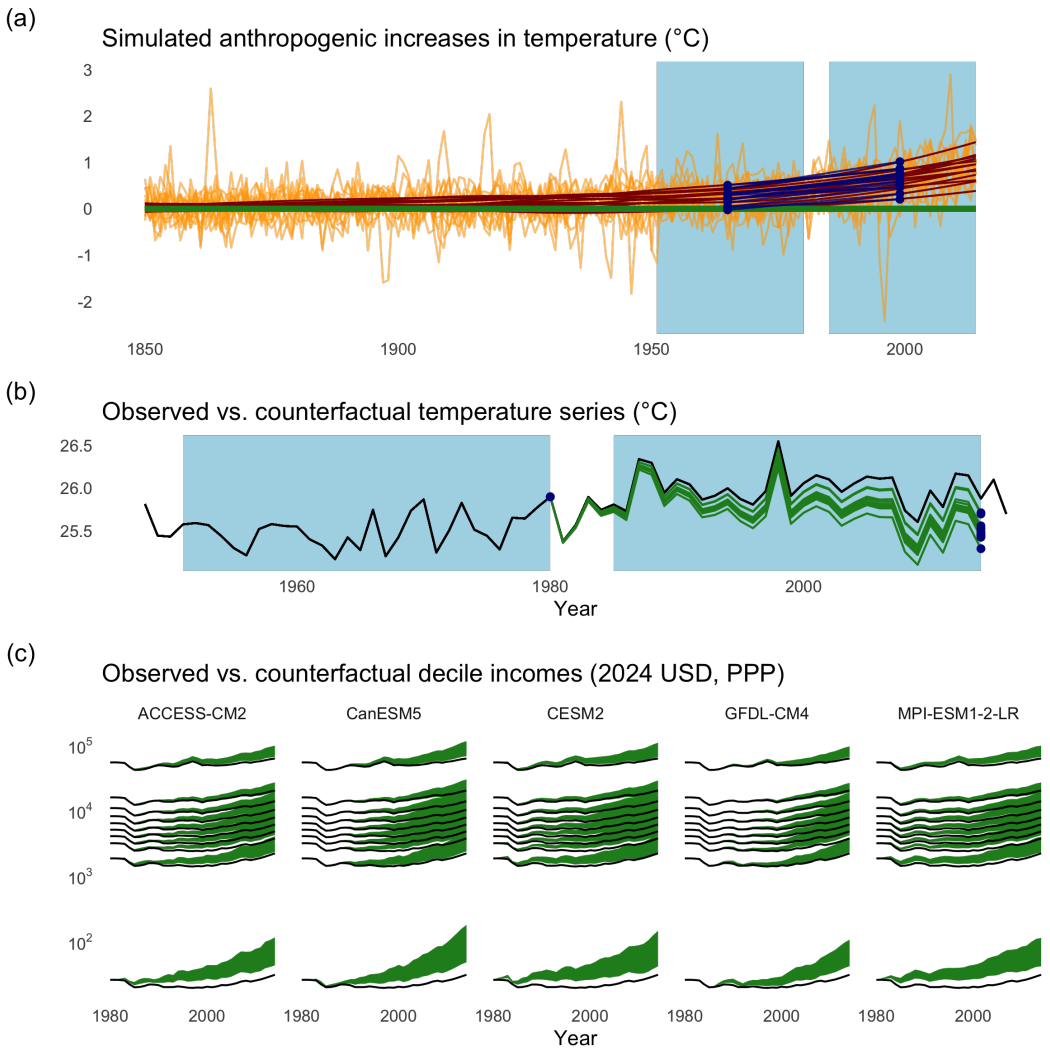
Our historical counterfactual analysis uses simulations from a category of experiments called "historical runs". For the CMIP6 generation of models, historical simulations begin at a pre-industrial baseline period in 1850 and run through to 2014. Among the forcing scenarios used in these historical experiments is one simply labeled "historical", which attempts to represent the impacts of influential forcings actually imposed on the climate system over this period, such as the deforestation of the Amazon rainforest and the eruptions of Krakatoa in 1883 and Mt. Pinatubo in 1991. Results from this historical run are required of all participating CMIP6 models since these "hindcasts" can be directly compared to observational data in order to evaluate model accuracy. Models collectively perform remarkably well at reproducing the historical record (Liu et al., 2021; Zelinka et al., 2020; Zhou et al., 2020). For this reason, studies which use CMIP data for the purpose of counterfactual analysis are advised to include the full available "multi-model ensemble" for experiments of interest (Knutti et al., 2010; Tebaldi & Knutti, 2007; van Vuuren et al., 2006).

A subset of these models also simulate a set of optional 'historical-natural' experiments which are defined identically except with anthropogenic forcings held fixed at pre-industrial levels. The differences between the two forcings by the same model holding all other variables fixed is then interpreted as the model's simulation of anthropogenic contributions to climate change. Comparisons between historical and historical-natural experiments are commonly used in the subfield of attribution science concerned with quantitatively mea-

[18] Natural forcings are distinct from sources of natural variability such as the ENSO cycle because they are external to coupled climate models.

Figure A6: Constructing counterfactuals holding anthropogenic forcings fixed at 1980 levels

Demonstration of the delta method for constructing bias-corrected counterfactuals using data for the Philippines as an example.



suring anthropogenic contributions to the intensity, frequency, or probability of weather events or trends.

For each climate model m and country i, we use an observed series T and the two model-specific historical series $\mathcal{T}^{hist}$ and $\mathcal{T}^{nat}$ to construct a counterfactual series $\widetilde{T}$ where anthropogenic forcings have been held fixed since the first year of the simulation period. The iterative process is summarized in Algorithm A2 and illustrated in Figure A6 using data from the Philippines as an example. In panel (a), differences between model-specific simulations of national temperature when including all forcings (orange) and when excluding anthropogenic forcings (green horizontal axis). Local regression fits are depicted in red. Light-blue shaded regions correspond to 30-year reference periods used to define temperature normals (dark blue dots) for the start and end of the simulation period 1980-2014.

In panel (b), The linearized difference between normals depicted as dark blue lines in (a) are subtracted from the observed temperature series (black) starting in 1980 to produce counterfactual temperature series (green). Each of these is interpreted as the temperature history that would have been realized had anthropogenic forcings been held fixed at 1980 levels according to a specific model of the climate system.

In panel (c), linearizing the difference between the climate normals defined at either end of the simulation period and subtracting the result from an observed temperature series implements the standard "delta method" for constructing bias-corrected climate counterfactuals. Counterfactual economic series for a random selection of five climate models are depicted in green and are constructed by iteratively applying the decile-level dynamic income responses depicted in Figure 4 to the observed and counterfactual temperatures depicted in (b). Their difference is interpreted as the change in income attributable to anthropogenic forcings since 1980. Green bands correspond to 90% confidence intervals generated from 500 bootstrap estimates of the response functions.

Algorithm A2: Constructing temperature counterfactuals holding anthropogenic forcings fixed at 1980 levels

Input: $\{\mathcal{T}^{hist}_{m,i,t}\}$,$\{\mathcal{T}^{nat}_{m,i,t}\}$,$\{T_{it}\}$

for $t^* \in \{1980, 2014\}$ **do**

$\overline{\mathcal{T}}^{hist}_{m,i,t^*} \leftarrow \frac{1}{30}\sum_{j=1}^{30} \mathcal{T}^{hist}_{m,i,t^*-j}$;

$\overline{\mathcal{T}}^{nat}_{m,i,t^*} \leftarrow \frac{1}{30}\sum_{j=1}^{30} \mathcal{T}^{nat}_{m,i,t^*-j}$;

$\overline{\delta}_{m,i,t^*} \leftarrow \overline{\mathcal{T}}^{hist}_{m,i,t^*} - \overline{\mathcal{T}}^{nat}_{m,i,t^*}$;

end

for $t = 1980$ **to** 2014 **do**

$\overline{\delta}_{m,i,t} \leftarrow (t - t_0)\frac{\overline{\delta}_{m,i,2014} - \overline{\delta}_{m,i,1980}}{2014-1980}$;

$\widetilde{T}^{nat}_{m,i,t} \leftarrow T_{it} - \overline{\delta}_{m,i,t}$;

end

Construct states $\{\overline{T}^{nat}_{m,i,t}\}$ by applying (1a) to $\{\widetilde{T}^{nat}_{m,i,t}\}$ then subtracting by $\widetilde{T}^{nat}_{m,i,1980}$;

Construct shocks $\{\widehat{\tau}^{nat}_{m,i,t}\}$ by applying (1a) and (1b) to $\{\mathcal{T}^{nat}_{m,i,t}\}$;

Output: $\{\widetilde{T}^{nat}_{m,i,t}\}$,$\{\overline{T}^{nat}_{m,i,t}\}$,$\{\widehat{\tau}^{nat}_{m,i,t}\}$

Algorithm A3: Constructing economic counterfactuals holding anthropogenic forcings fixed at 1980 levels

Input: $\{\widetilde{T}^{nat}_{m,i,t}\}$,$\{\overline{T}^{nat}_{m,i,t}\}$,$\{\widehat{\tau}^{nat}_{m,i,t}\}$, B bootstrap samples indexed b

Initializing series:

  for $b = 1$ **to** B **do**

    Specify desired simulation interval: $(t_0, t_1) \leftarrow (1980, 2014)$;

    Estimate $\mathcal{R}^{c}_{\tau \rightarrow Y}(h; \overline{T}, b)$ from (5);

    Specify maximum projection horizon H using persistence test from Section A2.2;

    Define IRFs $\mathcal{R}_{\tau \rightarrow Y}(h; \overline{T}, b) := \mathcal{R}^{c}_{\tau \rightarrow Y}(h; \overline{T}, b) - \mathcal{R}^{c}_{\tau \rightarrow Y}(h-1; \overline{T}, b)$;

    Collect IRF coefficients;

    $\{\widehat{\beta}_{1,h,b}, \widehat{\beta}_{2,h,b}, \widehat{\lambda}_{h,b}\}_{h=0}^{H}$;

    Define $\delta^{\tau}_{m,b,i,t} := \widehat{\tau}^{nat}_{m,i,t} - \widehat{\tau}_{it}$;

    Define $\delta^{\tau T}_{m,b,i,t} := \widehat{\tau}^{nat}_{m,i,t} \cdot \overline{T}^{nat} - \widehat{\tau}_{it} \cdot \overline{T}_{it}$;

    Define $\delta^{T}_{m,b,i,t} := \overline{T}^{nat}_{m,i,t} - \overline{T}_{it}$;

    Initialize $\delta^{Y}_{m,b,i,t} \leftarrow 0$;

    Initialize $\widetilde{Y}^{nat}_{m,b,i,t_0} := Y_{i,t_0}$

  end

Iteratively constructing series:

  for $t = t_0 + 1$ **to** t_1 **do**

    for $b = 1$ **to** B **do**

      for $h = 0$ **to** H **do**

        $\delta^{Y}_{m,b,i,t+h} \leftarrow \delta^{Y}_{m,b,i,t+h} + \widehat{\beta}_{1,h,b} \cdot \delta^{\tau}_{m,b,i,t} + \widehat{\beta}_{2,h,b} \cdot \delta^{\tau T}_{m,b,i,t} + \widehat{\lambda}_{h,b} \cdot \delta^{T}_{m,b,i,t}$;

      end

      $\widetilde{Y}^{nat}_{m,b,i,t} \leftarrow \left(1 + \log(\frac{Y_{it}}{Y_{i,t-1}}) + \delta^{Y}_{m,b,i,t}\right) \widetilde{Y}^{nat}_{m,b,i,t-1}$

    end

  end

Output: $\widetilde{N}^{nat}_{m,b,i,t}$

Figure A7: Anthropogenic growth incidence relative to 1980 baseline by climate model

